



SMALL SCALE GROUPED HYDROPOWER PROJECT BY PHONGSUBTHAVY GROUP SOLE CO., LTD



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1 PROJECT DETAILS

1.1 Summary Description of the Project

The grouped VCS project aims to promote the implementation of small-scale hydropower plants in the Lao People's Democratic Republic (Lao PDR). The project proponent will contribute to promote the implementation of Small Hydropower Plants in the Lao PDR, wherein the generated electricity will be supplied to national grid of Vietnam.

The proposed grouped VCS project, is promoted by Phongsubthavy Group Sole Co., Ltd., wherein the project developed by subsidiary company of Phongsubthavy Group Co., Ltd, Nam Boak Hydropower Sole Co., Ltd, Green Star Co., Ltd and/or other project implementor will be included as an instance.

Small-scale hydropower generation complies with the framework of the energy sector policies of Lao PDR in aspects related to renewable energy electricity generation and technological innovation. There is no regulation in Lao PDR mandating the implementation of small hydropower-based projects for electricity generation. Moreover, there are no promotion tools for this technology and does not exist any regulation or mandate to develop small hydropower projects in the country.

The instances under grouped VCS project will be located in host country Lao PDR and generated electricity will be supplied to Vietnam National grid through direct transmission line.

The first actual instance under grouped VCS project is proposed 8.1 MW Nam Boak Hydro Power project located on Nam Mo River in the Muang Mok district, Xieng Khoang province, Lao PDR. It is run-of-river (RoR) small scale hydro project with reservoir of about 0.007 km², which has storage capacity of 37.515 m³. The power generated by the project will be supplied to the regional grid consisting of Vietnam Power Grid directly without connecting to the Lao Power Grid. The proposed project activity is a run of river project but has created a very small reservoir for smooth operation of the project activity.

The proposed installed capacity of the project is 8.1 MW comprising 3 turbines with individual installed capacity 2.7MW each, with estimated annual power generation 28.84 GWh¹. The generated electricity from the project activity will be supplied to the regional grid connected to Vietnam National grid. The project activity is expected to be commissioned dated 01 October 2025.

The purpose of the project activity is to produce renewable energy using hydro power for supplying to grid. In the absence of the project activity equivalent amount of electricity would have been generated by the operation of grid connected fossil fuel based power plants. The Project activity

¹ Detailed Project Report

thus reduces the anthropogenic emissions of greenhouse gases (GHGs) into the atmosphere associated with the equivalent amount of electricity generation from the fossil fuel based grid connected power plant.

The baseline scenario is that the electricity delivered to the grid by the project activity would have otherwise been generated by the operation of grid-connected power plants and by the addition of new generation sources into the grid.

The estimated annual average and the total CO₂e emission reduction by the first instance over the first renewable crediting period of 7 years are expected to be 20,615 tCO₂e and 144,305 tCO₂e respectively.

1.2 Audit History

Audit type	Period	Program	Validation/verification body name	Number of years
<i>Validation</i>	--	VCS	TBA	-
	...			

1.3 Sectoral Scope and Project Type

Sectoral scope	01 – Energy Industries (Renewable -/non-renewable sources)
Project activity type	Type-I Renewable Energy Project

1.4 Project Eligibility

1.4.1 General eligibility

The project activity is 8.1MW small scale hydro power project located in Laos PDR a least development country as per United Nation list of LDC. As per para 3.10.1 of VCS Standard V4.7, the project activity falls under the category of “project” as it has estimated annual emission reduction potential less than 300,000 tonnes of CO₂e per year. The project activity is eligible to use small scale approved methodology AMS-I.D Version-18.0. Therefore, in line with section 2.1 of VCS Standard version-4.7 the project activity is eligible to use VCS programme of VERRA.

As per para 3.8.1 of VCS Standard version 4.7, Non-AFOLU projects shall complete validation within two years of the project start date. The project activity will be listed with VERRA for public comments after November 2024 prior to commissioning of the project activity i.e. 01/10/2025 which is expected start date, hence project activity complies with the VCS listing/validation completion requirements.

As per para 9 of TOOL 20, Assessment of Debundling for Small-Scale Project Activities” V4.1, A proposed small-scale project activity shall be deemed to be a debundled component of a large project activity if there is a registered small-scale CDM project activity or an application to register another small-scale CDM project activity:

- a) With the same project participants;
- b) In the same project category and technology/measure; and
- c) Registered within the previous 2 years; and
- d) Whose project boundary is within 1 km of the project boundary of the proposed small- scale activity at the closest point.

The proposed hydro power project commissioned is yet to be commissioned and PP has no other small-scale or large-scale project commissioned within 1 km of the project boundary, hence proposed project activity is not a fragmented part of the larger project activity.

VCS Scope		Justification
2.1.1	The scope of the VCS Program includes: a) The seven Kyoto Protocol greenhouse gases. b) Ozone-depleting substances (ODS). c) Project activities supported by a methodology approved under the VCS Program through the methodology development and review process. d) Project activities supported by a methodology approved under an approved GHG program, unless explicitly excluded (see the Verra website for exclusions). e) Jurisdictional REDD+ programs and nested REDD+ projects as set out in the VCS Program document <i>Jurisdictional and Nested REDD+ (JNR) Requirements</i>.	a) The project activity aims to reduce CO₂e emission reductions are GHG gases identified under Kyoto Protocol b) Not applicable c) Not applicable d) The project activity uses CDM approved small scale methodology AMS-I.D, the CDM GHG programme is approved by VCS. Hence criterion is met. e) Not applicable
2.1.2	The scope of the VCS Program excludes projects that can reasonably be assumed to have generated GHG emissions primarily for the purpose of their subsequent reduction, removal, or destruction.	The project activity has not been implemented to generate GHG emissions for the purpose of their subsequent reduction, removal or destruction. Hence criterion is met.
2.1.3	The VCS Program also excludes the project activities under the circumstances indicated in	The project activity is 8.1MW small scale hydro power project

	Table 1 of clause 2.1.3 of the VCS Standard version-4.7	located in Laos PDR a least development country ² as per United Nation list of LDC. In line with guidance under table 1, the project activity is not under exclusion category.
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1.4.2 AFOLU project eligibility

The project is not an AFOLU Project. Thus, this section is not applicable.

1.4.3 Transfer project eligibility

The project activity is not a transfer project or a CPA seeking registration. Thus, this section is not applicable.

1.5 Project Design

Indicate if the project has been designed as:

- ☐ Single location or installation
- ☐ Multiple locations or project activity instances (but not a grouped project)
- ☒ Grouped project

1.5.1 Grouped project design

For inclusion of a new project activity instance, the project proponent shall ensure that it meets the following eligibility criteria:

S No	Criterion	How the new project activity instances comply	Evidence
1	The geographical boundary of the PAI including any time-induced boundary consistent with the geographical boundary set in the grouped project	Each PAI will be located in host country Lao PDR.	The same will be checked with physical address along with geo-coordinates of the project site and nodal agency approval for establishment and

² <https://www.un.org/en/conferences/least-developed-countries#:~:text=The%2046%20countries%20currently%20on,%2C%20Kiribati%2C%20Lao%20People's%20Dem.>

			operation of the instances. PAI-1: The proposed 8.1 MW Nam Boak Hydro Power project located on Nam Mo River in the Muang Mok district, Xieng Khoang province, Lao PDR. Hence is within geographical boundary of Lao PDR.
2	Avoidance of double counting Conditions to confirm that PAIs are neither registered as VCS project, included in another registered project/ grouped project/ CDM/GS PoAs or in any other GHG programme, nor the project activities that have been deregistered	The PAI is not part of any other PoA or will not be registered as individual project under any other emission reduction scheme or programme, or deregistered CDM/VCS/GS activity,	Unique identification of PAI and check with other GHG programme database and a declaration from project implementor. PAI-1: The proposed project activity is still under construction and is not registered or seeking registration under any other GHG programme as an individual project or as an instance. Hence, criteria met.
3	Applicable Target Group for PAI Inclusion	PAI belongs renewable energy-based grid electricity generation. applying hydro Energy based technology measure. The PAI supplies electricity to the national/regional grid.	At least one of the following: • PAI implementation technical design information or feasibility report or Detailed Project Report • Power Purchase Agreement • Grid Evacuation approval/ agreement

			Power Purchase agreement PAI-1: The proposed project activity is a 8.1 MW Hydro energy based power generation project, the same is checked from Feasibility Study Report, prepared by third party engineering company. Hence, criterion is met.
4	Specification of the technology/measure, such as the level and type of service, as well as performance specification based on, inter alia, testing/certification.	The installed capacity of the PAI will be less than or equal to 15MW of hydro Energy technology measure to supply electricity to the grid. Electricity produced by the project activity will be monitored at the point it is exported to the grid.	The supporting technology description (PAI implementation technical design information or feasibility report or Detailed Project Report) or commissioning certificate. PAI-1: The installed capacity for the proposed project is 8.1MW, which is less than 15MW, the same has been checked with FSR. Hence criterion is met.
5	Conditions to check the start dates of PAIs through documentary evidence.	The PAIs having started after the start date of the first PAI or completed validation within 2 years from start date.	PAI start date evidence i.e. date when PAI commissioned. PAI-1: The proposed first instance i.e. Nam Boak 8.1MW hydro power project is expected to be commissioned in October 2025. Further, this is first instance, hence criterion is met.
6	Conditions to ensure compliance with the	Each PAI will be small scale hydro energy based power	Each PAI under the grouped project will have

	applicability of the applied methodologies, the applied standardized baselines and the other applied methodological regulatory documents.	generation project, the following Type I small-scale methodology is applied: AMS-I.D (Version 18.0)	to meet the applicability condition of AMS-I.D. PAI-1: The proposed project activity is meets all the eligibility conditions of applied approved methodology AMS-I.D as established in section 3.2. Hence, criterion is met.
7	Environmental Assessment and Local Stakeholder Consultation	PAI Implementer to conduct an environmental assessment if required as per host country regulation and have identified and appraised any affected local stakeholders.	If applicable Environmental Assessment Report, including Preliminary Environmental and Social Assessment, prepared for submission for either regulatory approval or for any lender's consideration of approval for project finance. If applicable, evidence of any local consultation process completed, and how any comments received were considered. PAI-1: The proposed project activity has achieved the EIA approval / Environment Compliance Certificate (ECC) No.899/DONRE.XKH dated 20/05/2021. The PP has conducted local stakeholder along with Initial Environmental Examination (IEE) study. Hence criterion is met.

8	Are subject to the baseline scenario determined in the project description for the specified project activity and geographic area.	All new PAIs will be implemented only in regions within the geographic borders of Lao PDR and the generated electricity to be supplied to national grid of Vietnam subject to the same baseline scenario determined in Section 3.4 in this Project Description.	The baseline for each PAI will be Vietnam national grid and will be confirmed for each PAI through PPA. PAI-1: The proposed project activity will supply the net generated electricity to National grid of Vietnam through long term PPA. Hence, criterion is met.
9	Have characteristics with respect to additionality that are consistent with the initial instances for the specified project activity and geographic area.	All PAIs will use the activity method for demonstration of additionality. Step 1: Regulatory Surplus There is no government mandated programme or policy in host country for implementation of hydro power project or selection of technology for power generation. Step 2: Positive List The project activity is not in positive list. Project Method: To demonstrate additionality by applying provisions from one of the following conditions as applicable to a specific PAI: • Additionality based on the “TOOL21: <u>Demonstration of additionality of small-scale project activities</u>”.	The specific PAI shall demonstrate additionality described in section 4.5 of this PD, and including with all relevant supporting information to transparently demonstrate additionality. PAI-1: The proposed project activity is not a financially attractive option in comparison to benchmark as established in section 3.5. Hence, criterion is met.

10	Where a capacity limit applies to a project activity included in the project, no PAI shall exceed such limit. Further, no single cluster of project activity instances shall exceed the capacity limit, determined as follows: Each device that exceeds one percent of the capacity limit shall be identified. Such instances shall be divided into clusters, whereby each cluster is comprised of any system of instances such that each instance is within one kilometer of at least one other instance in the cluster. Instances that are not within one kilometer of any other instance shall not be assigned to clusters. None of the clusters shall exceed the capacity limit and no further project activity instances shall be added to the project that would cause any of the clusters to exceed the capacity limit.	Each PAI under the grouped project shall have installed capacity less than or equal to 15MW.	The installed capacity for the proposed project activity is 8.1MW which is less than 15MW. The same is checked from FSR and EPC contract, There are no other project by the project proponent within a kilometer from the project boundary. Hence the criterion is met.
Eligibility conditions specific to inclusion of New PAIs			
11	The new PAI must occur within one of the	The PAIs under grouped project will be developed and	Each PAI under grouped project will be

	designated geographic areas specified in the project description	commissioned within the geographical boundary of host country Lao PDR.	implemented within host country Lao PDR. The same can be checked with FSR, nodal agency approvals and commissioning certificates. For the current PAI the FSR, EIA Approval, PPA shared with VVB.
12	The new PAI must comply with at least one complete set of eligibility criteria for the inclusion of new project activity instances. Partial compliance with multiple sets of eligibility criteria is insufficient.	The eligibility criteria set must be complied by the PAIs at all points in time throughout the crediting period. If at any point, it is found that the PAI is not complying with the set of eligibility criteria, then it may be excluded from the crediting for the duration of the monitoring period, until it is able to demonstrate the compliance of the eligibility criteria.	The MR will act as the documentary compliance summary. However, for each specific eligibility criteria (as laid down above), the supporting evidence will be provided by the PP.
13	The new PAI must be included in the monitoring report with sufficient technical, financial, geographic and other relevant information to demonstrate compliance with the applicable set of eligibility criteria and enable sampling by the validation/verification body.	The PAIs will include all the technical, financial, geographic and other relevant information related to the compliance of the applicable set of eligibility criteria & also enable sampling by the VVB.	The MR for the specific monitoring period will act as the supporting evidence for the stated condition/requirement.
14	The new PAI must be validated at the time of	The PAI(s) shall be part of the monitoring, and therefore,	The Verification Report based on the VCS

	verification against the applicable set of eligibility criteria.	will be validated against the criteria specified, by an VCS accredited/approved VVB.	template, shall be issued by the approved VVB. This will act as the supporting evidence against this requirement.
15	The new PAI must have evidence of project ownership, in respect of each project activity instance, held by the project proponent from the respective start date of each project activity instance (i.e., the date upon which the project activity instance began reducing or removing GHG emissions).	The PAI related evidence on the project ownership will be in the form one or more of below. 1. Concession agreement (CA) 2. Construction approval/Commissioning certificate 3. Power Purchase Agreement (PPA) 4. Commissioning certificate	The EIA approval, PPA, Concession agreement and commissioning certificate to be checked for the PAIs to be included.
16	The new PAI must have a start date that is the same as or later than the grouped project start date.	The Grouped project is under planning stage right now, and therefore, the PAIs will have a start date later than the grouped project start date.	This can be verified through the commissioning date of individual instance, wherein each PAI can be verified for its start date.
17	The new PAI must be eligible for crediting from the start date of the instance through to the end of the project crediting period (only).	The PAI will be only eligible for the crediting from its start date through to the end of the project crediting period (only).	1. The start date of the PAI can be checked through the commissioning certificate. 2. The end date of the grouped project crediting shall be referenced from the VCS Project Registry webpage. 3. The overlap of monitoring periods may be checked through the previous monitoring reports (available publicly

			on VCS project registry webpage for those periods).
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1.6 Project Proponent

Organization name	Phongsubthavy Group Sole Co., Ltd
Contact person	Mone Nouansyvong
Title	Director
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Email	Mone.Nouansyvong@phongsupthavy.la

1.7 Other Entities Involved in the Project

Organization name	Nam Boak Hydropower Sole Co., Ltd
Role in the project	Legal Owner and developer
Contact person	Mr. Phongsavath Senaphuan
Title	Director
Address	New Nongbuathong – Nongbuek Road, House No 88, Nongbuektai Village, Sikhodtabong District, Vientiane Capital Lao PDR; PO Box: 9889
Telephone	Tel: (856-21) 618000, 618088 Fax: (856-21) 618077.
Email	phong@phongsub.com

Organization name	Green Star Co., Ltd
Role in the project	Implementor

Contact person	Mone Nouansyvong
Title	Director
Address	New Nongbuathong – Nongbuek Road, House No 88, Nongbuektai Village, Sikhodtabong District, Vientiane Capital Lao PDR; PO Box: 9889
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Organization name	Rtayte Pte Ltd
Role in the project	Consultant
Contact person	Mr. Phool Chand
Title	Director
Address	3 Shenton Way, #15-05, Shenton House, Singapore – 068805
Telephone	+65 8380 1549
Email	Carbontrade@rtayte.com

1.8 Ownership

The following evidences can verify the ownership of the Project and its right to use:

1. Concession agreement (CA)
2. Construction approval/Commissioning certificate
3. Power Purchase Agreement (PPA)
4. EIA approval/Environment Compliance Certificate (ECC) No.899/DONRE.XKH dated 20/05/2021.

The project activity is developed and owned by Nam Boak Hydropower Sole Co., Ltd , a subsidiary of Phongsubthavy Group Sole Co., Ltd (PSTG). Nam Boak Hydropower Sole Co., Ltd has authorised Phongsubthavy Group Sole Co., Ltd to develop project under VCS and to claim carbon credits/VER, thereby Phongsubthavy Group Sole Co., Ltd has ownership of generated VERs from VCS grouped project activity for entire crediting period. All implementor under the grouped project will authorize Phongsubthavy Group Sole Co., Ltd to develop VCS project and claim emission reduction for entire crediting period through an agreement.

1.9 Project Start Date

Project start date	01-October-2025
Justification	Expected date of commissioning of the first instance i.e. 8.1MW Nam Boak hydro power project

1.10 Project Crediting Period

Crediting period	☒ Seven years, twice renewable ☐ Ten years, fixed ☐ Other (state the selected crediting period and justify how it conforms with the VCS Program requirements)
Start and end date of first or fixed crediting period	01-October-2025 to 30-September-2032 (inclusive of both start and end date)

1.11 Project Scale and Estimated GHG Emission Reductions or Removals

- ☒ < 300,000 tCO₂e/year (project)
☐ ≥ 300,000 tCO₂e/year (large project)

Calendar year of crediting period	Estimated GHG emission reductions or removals (tCO ₂ e)
01-Oct-2025 to 31-Dec-2025	5,196
01-Jan-2026 to 31-Dec-2026	20,615
01-Jan-2027 to 31-Dec-2027	20,615
01-Jan-2028 to 31-Dec-2028	20,615
01-Jan-2029 to 31-Dec-2029	20,615
01-Jan-2030 to 31-Dec-2030	20,615
01-Jan-2031 to 31-Dec-2031	20,615
01-Jan-2032 to 30-Sep-2032	15,419
Total estimated ERRs during the first or fixed crediting period	144,305

Total number of years	7
Average annual ERRs	20,615

1.12 Description of the Project Activity

The proposed grouped VCS project will involve commissioning and operation of small hydro power plants in Lao PDR, under category-I, “Renewable energy projects”, with installed capacity under or equal to 15 MW and in compliance with the following criteria, eligible under the applicable small scale methodology AMS-I.D “Grid connected renewable electricity generation” version 18:

Small hydropower plants in the Lao PDR:

- Supplying electricity to the grid; or
- Supplying electricity to an identified consumer facility via national/regional grid through contractual arrangement;

The Run of river small hydro power plants, may involve small reservoir for water storage. The river flow will be diverted from the natural water course, maintaining the minimum ecological water flow to the natural river bed. After the generators and turbines, the water flow will be charged back in the river bed.

The proposed first instance is 8.1 MW Nam Boak Hydro Power Project is located on Nam Mo river in the Muang Mok district, Xieng Khoang province, Lao PDR. The proposed small scale hydro project is a run-of-river (RoR) project has created reservoir covering area 0.007 km².

The proposed project is new 8.1MW hydropower plant is a greenfield project to be commissioned as a standalone project at a site having no existing hydropower project.

The power generated by the project will be supplied to the national grid of Vietnam. According to the Feasibility Study Report (FSR), the annual net electricity supply will be 28.84 GWh, with PLF 40.64%. The proposed installed capacity of the project is 8.1MW; the area of reservoir at full water level is 0.007 Km². Therefore, the power density (PD) = $8,100,000 / 7,852 = 1,031.58 \text{ W/m}^2$ which is greater than 10 W/m². The project activity is expected to be commissioned dated 01st October 2025.

The power generated shall be fed using PSG’s gird to Vietnam Electricity (EVN) grid at 115 kV grid transmission line through Nam Xao No. 3 35/115kV step up sub-station.

The Francis turbine that will be installed as a part of the project activity is a horizontal flow double jet turbine, water flows along the tangent to the path of the runner. Nozzles direct forceful streams of water against a series of spoon-shaped buckets mounted around the edge of a wheel. Each bucket reverses the flow of water, leaving it with diminished energy. The resulting impulse spins the turbine. The buckets are mounted in pairs, to keep the forces on the wheel balanced, as well as to ensure smooth, efficient momentum transfer of the fluid jet to the wheel. The Pelton

turbine is most efficient in high head applications. This technology is generally considered to be environmentally safe and sound technology and is used for hydro power generation worldwide. The project activity does not involve the technology transfer from any country. The expected annual Net Electricity generation to be supplied to grid is expected to be 28,840 MWh. The lifetime of turbines installed is around 25 years as per manufacturer and average Industry standard with required maintenance and repair. The PLF is estimated as 40.64% as per feasibility study report prepared by third party engineering consultant.

Technical Specification of Turbine

Rated Net Head	131.61 m
Design Flow	6.92 m ³ /s
Make	Czech
Rated Power	2,700 kW
Turbine Type	Horizontal Francis
Number of Turbine	2
Operational lifetime	25 years

Technical Specification of Generator

Make	Czech
Capacity	3 x 2,700 kW
Voltage Output	6.3 kV
Frequency	50Hz
Configuration	Horizontal
Type	Synchronous SESR, Brushless

1.13 Project Location

The grouped project activity will have project boundary as physical boundary of host country Laos PDR. The proposed first instance i.e. Nam Boak Hydro Power Project is located on Nam Mo river in the Muang Mok district, Xieng Khoang province, Lao PDR. Dam and powerhouse's location are as follows.

	Coordinates	
	Latitude	Longitude
Dam	19°01'56.60"N	104°02'00.10"E
Powerhouse	19°01'53.9"N	104°02'28.6" E



Fig: Project Site.

The KML file is provided for first instance.

1.14 Conditions Prior to Project Initiation

The project activity supply electricity to national grid of Vietnam and in absence of proposed project the equivalent amount of electricity would have continued to be generated in grid mix power plants, which is also the baseline scenario.

1.15 Compliance with Laws, Statutes and Other Regulatory Frameworks

There are no laws and regulations governing the use of Hydro-electric power as electricity in Laos. The project is a voluntary effort by the project proponent. The government promotes use of clean fuel. There is no specific concern made on Hydro Power project from the host country laws and regulations.

1.16 Double Counting and Participation under Other GHG Programs

1.16.1 No Double Issuance

Is the project receiving or seeking credit for reductions and removals from a project activity under another GHG program?

☐ Yes

☒ No

1.16.2 Registration in Other GHG Programs

Has the project registered under any other GHG programs?

☐ Yes ☒ No

Is the project active under the other program?

☐ Yes ☒ No

1.16.3 Projects Rejected by Other GHG Programs

Has the project been rejected by any other GHG programs?

☐ Yes ☒ No

1.17 Double Claiming, Other Forms of Credit, and Scope 3 Emissions

1.17.1 No Double Claiming with Emissions Trading Programs or Binding Emission Limits

Are project reductions and removals or project activities also included in an emissions trading program or binding emission limit? See the *VCS Program Definitions* for definitions of emissions trading program and binding emission limit.

☐ Yes ☒ No

1.17.2 No Double Claiming with Other Forms of Environmental Credit

Has the project activity sought, received, or is planning to receive credit from another GHG-related environmental credit system? See the *VCS Program Definitions* for definition of GHG-related environmental credit system.

☐ Yes ☒ No

1.17.3 Supply Chain (Scope 3) Emissions

Do the project activities specified in Section 1.12 affect the emissions footprint of any product(s) (goods or services) that are part of a supply chain?

☐ Yes ☒ No

If yes:

Is the project proponent(s) or authorized representative a buyer or seller of the product(s) (goods or services) that are part of a supply chain?

☐ Yes ☒ No

If yes:

Has the project proponent(s) or authorized representative posted a public statement on their website saying, “Carbon credits may be issued through Verified Carbon Standard project [project ID] for the greenhouse gas emission reductions or removals associated with [project proponent or authorized representative organization name(s)] [name of product(s) whose emissions footprint is changed by the project activities].”

☐ Yes

☒ No

The project activity doesn't claim emission reduction due to supply chain of goods, as the project is a hydro energy based power generation project wherein emission reduction is only claimed due to avoidance of fossil intensive grid electricity. Hence, scope 3 emission double counting is not applicable to project activity

1.18 Sustainable Development Contributions

The project is expected to contribute 5 SDGs which are SDG 7, 8, 5, and 13.

- SDG 7 Energy: The project contributes SDG Target 7.2.1 “By 2030, increase substantially the share of renewable energy in the global energy mix” by the utilization of hydropower as a renewable energy source. The project activity is expected to inject 28,840 MWh/year to grid.
- SDG 8 Economic Growth: The project creates direct and indirect employment opportunities during construction and operation phases, so it contributes to SDG Target 8.5.2 “By 2030, achieve full and productive employment and decent work for all women and men, including for young people and persons with disabilities and equal pay for work of equal value”. The project activity is expected to create 45 regular employment during operation.
- SDG 5 Women employment: The project contributes SDG Target 5.1 “End all forms of discrimination against all women and girls everywhere” by providing employment to female and contributing to Women employment and financial independence. The project activity is expected to have around 30-40% women employment during operation.
- SDG 13 Climate Change: The project produces clean renewable energy by diminishing CO₂ emissions. Therefore, it contributes SDG Target 13.2.2 “Total greenhouse gas emission per year”. The project activity expected to reduce 20,615 tCO₂e per year.

1.19 Additional Information Relevant to the Project

1.19.1 Leakage Management

As per applied approved methodology no leakage emission relevant to project activity.

1.19.2 Commercially Sensitive Information

No commercially sensitive information has been excluded from the public version of the project description.

1.19.3 Further Information

Not applicable.

2 SAFEGUARDS AND STAKEHOLDER ENGAGEMENT

2.1 Stakeholder Engagement and Consultation

2.1.1 Stakeholder Identification

Stakeholder Identification	The local stakeholders are those who face the immediate effect due to the project activities, which involves effect on the local environment, social life and economics. They can be within the boundaries of the village, district, state or nation. The local stakeholder consultation process was developed to capture the feedback of all relevant stakeholders with details of grouped VCS project and potential impacts. The relevant stakeholders being identified by the project proponent as employee, vendor, villagers nearby and NGOs.
Legal or customary tenure/access rights	The grouped VCS project activity is developed, commissioned and operated by Nam Boak Hydropower Sole Co., Ltd and a subsidiary of Phongsubthavy Group Sole Co., Ltd. is a legal owner of the project. The land used for construction of Nam Boak 8.1 MW hydro power project belongs to Nam Boak Hydropower Sole Co., Ltd (NBHP) and there is no conflict with any other stakeholders.
Stakeholder diversity and changes over time	The stakeholders for the project are from different category including employee, villagers, local leaders, vendors and NGOs. Based on project type no change in stakeholders is envisaged overtime.
Expected changes in well-being	The grouped VCS project will contribute in improving the environmental condition in the region of by generating emission free electricity using renewable resource i.e. hydro power. The project

	activity generated employment for skilled and unskilled workers due to project implementation and operation phase. The project activity supplies net electricity generated to grid, thereby results in emission reduction due to displacing equivalent electricity from grid dominated by fossil fuel based thermal power plants.
Location of stakeholders	The relevant stakeholders are the nearby villagers and employees. As the land used for implementation of the project belongs to PSG Group, hence, there is no conflict with any other stakeholders. The project is run of the river project; hence, project has no impact on stakeholders' field nearby or by other means.
Location of resources	The project instances under the grouped VCS project are run of the river projects, the first instance utilizes the water resources from Nam Mo river in the Muang Mok district, Xieng Khoang province, Lao PDR. The water diverted for project will be discharged to same river stream and between there is no habitation impacted due to the same. Hence, the project does not impact customary right to access water resources from the river for people living in nearby villages or downstream of the river.

2.1.2 Stakeholder Consultation and Ongoing Communication

Date of stakeholder consultation	13-December-2024
Stakeholder engagement process	The PP has identified relevant stakeholders as nearby villagers, nodal agencies, employees and vendors. The invitation letters being sent to identified stakeholders well in advance dated 15 November 2024 to attend the meeting and provide their inputs/feedback". The identified stakeholders have been informed by publishing invitation in "Vientiane Mai" a local newspaper and also by public notice in relevant places. The local villagers and the office bearers were also invited personally by a representative of the project proponent. The comments were invited in an open and transparent manner in the physical meeting by facilitating with a presentation on the proposed project activity.

	The physical meeting took place on 13th Dec 2024 at Mok district meeting hall. A total of 51 people attended the meeting, representing the following groups: 1. Local people, communities, and representatives expected to be directly or indirectly affected by or interested in the project 2. Local policymakers and representatives of local authorities 3. Local non-governmental organizations (NGOs) and Women Groups focusing on topics relevant to the project or collaborating with communities likely to be affected by the project. The stakeholder meeting organised at grouped project level and information details of instances shared with the stakeholders. The key issues discussed and question/concerns raised by attendee along with answers to them were recorded in minutes of meeting.
Consultation outcome	The local stakeholder's consultation meeting for the grouped VCS project was planned and conducted at Mok district meeting hall, wherein participation was expected from local stakeholders including NGOs, villagers, vendor and employees of project and any other entity who concerns itself with the actual project instances. The agenda of the meeting was to explain about the planned VCS grouped project and invite local stakeholders to understand about the project instances, submit their grievances, and raise their concerns, wherever required. Below are the steps followed during stakeholder consultation meeting. The meeting began with introductions of the objective and agenda of the meeting and participants by the meeting co-organizer, Xiengkhouang Provincial Natural Resources and Environment Department. The main purpose of the meeting was to discuss the project's non-technical summary and Environmental and Social Management and Monitoring Plan (ESMMP). Additionally, the potential environmental impacts of the project were reviewed, along with the measures the investor, Phongsubthavy Group Sole Co., Ltd will take to address these issues. <i>The meeting was started with the chairmanship of Mr. Vilaiphone Manivong, Deputy Director of the Natural Resources and</i>

	<i>Environment Department, to preside over the meeting and Mr. Somchai Boupphasanh, Vice District Governor, who was co-chairman of the meeting. All the participants have supported the proposal, accepted and consented the chairman and co-chairman to preside over the meeting. The meeting started with a brief introduction about Phongsubthavy Group Sole Co., Ltd, Mr. Vilaiphone Manivong gave the idea about the project to all the stakeholder setting the stage for an interactive session with the stakeholders.</i> While delivering the welcome address, Mr. Vilaiphone Manivong highlighted the purpose of the project to generate clean energy by using the hydro power and providing the energy to the grid. He explained the importance of the proposed project and allayed fears of any environmental pollution concerns due to the project activity. He added that the project would add impetus to the socio-economic fabric of the area and help in sustainable development, as promulgated by the VCS standard. The attendees of the meeting have expressed their satisfaction on the proposed project. There were no negative comment or concern raised by stakeholders during consultation process.
Ongoing communication	It was also informed to stakeholders attended the meeting, anyone having complains/feedback regarding the project during operation or during construction may contact site in-charge or project manager. The PP has provided email IDs of concerned persons at site and for escalation if needed. Any concern raised will be initially taken up by team at site, however if no satisfactory solutions provided the stakeholder can escalate the same to General Manager, the concerns raised will be taken and resolved within agreed time frame as reasonable. As project activity is commissioned, changes in technology or infrastructure are not envisaged, hence, regular meeting will not be needed, however if any concern/feedback received will be discussed and if require stakeholder will be invited again to discuss and take necessary action as may be needed.
Stakeholder input	➤ Do the GHGs cause global working? Good or Bad or What are advantage and disadvantage? Question asked by a presentative of the provincial department of energy and mine, and responses by department of climate change, ministry of natural resources and environment, Xiengkhouang

provincial department of natural resources and environment and Phongsubthavy Group Sole Co., Ltd are as follows.

Yes, the increase of the GHGs in Earth's atmosphere is a primary factor contributing to the rise in global temperatures due to the increased GHGs form greenhouse effects that trap the heat derived from sunlight/solar radiation that pass through and reflected from the earth surface.

Advantage or good points include increase plant growth and agriculture production in cold region, reduced harshness and mortality (living things) in the winter, and improved navigation in the artic areas, among others.

Disadvantage or bad poor effects include melting ice caps and glaciers, sea level rise and floods, climate change and extreme weather and its consequent effects, loss of biodiversity and ecosystems, threats to agricultural and forestry production and food security, drought and water scarcity, health impacts economic losses, among others. In general, the negative consequences far outweigh these potential advantages.

Q1. I strongly support the carbon credits project and would like to take immediate action and cooperation to protect forest in the upper stream or watershed, reduce forest encroachment to secure water and environment for electricity generation by hydropower project.

Question asked by a representative from the district public work and transport office, and responses by Xiengkhouang provincial department of natural resources and environment, vice district governor and Phongsubthavy Group Sole Co., Ltd are as follows.

We are working together, public, private sector local authorities to create and implement watershed including forest resources management plan. In addition, we need to implement land allocation programme, especially land allocation for agricultural production.

Q2. I agree with the project development, but would like to propose to the relevant authorities the dissemination of the Environmental and Social Management and Monitoring Plan (ESMMP) and carbon credits projects to wider communities, to raise awareness and encourage meaningful participation.

Question asked by a representative from the district agriculture and forestry office, and responses by Xiengkhouang provincial department of natural resources and environment and Phongsubthavy Group Sole Co., Ltd are as follows.

Yes, the dissemination of the ESMMP including carbon credits will be implemented as it is in the plan.

Q3. I would like to make a comment on benefits and stakeholders, in addition to outlining the advantages and stakeholders, it is important to clearly communicate how local

communities or villagers will benefit from the projects and carbon credits, as well as how local residents can get involved in these initiatives.

Question asked by a representative of the provincial assembly and responses by Phongsubthavy Group Sole Co., Ltd are as follows.

Yes, we are making every effort to clearly communicate the benefits of the project. For example, the proposed project will employ local people as per the requirements and eligibility during the construction and operation phase and will continue to prioritize hiring local residents for any future employment needs.

Q4. I would like to make a comment on the monitoring. I would like to propose to increase the frequency and involvement of authorities in monitoring the implementation of the ESMMP. For example, monitoring could be increased to four times a year, with at least one monitoring session led by a delegation from the district leadership.

Question asked by a representative of the provincial assembly and responses by Phongsubthavy Group Sole Co., Ltd are as follows.

Yes, for regular monitoring, we can conduct it four times a year, including a monitoring mission by district leadership, particularly during the construction phase. Additionally, we will carry out emergency monitoring to address any urgent issues as they arise. These measures should be sufficient.

Q5. I would like to raise a concern regarding another issue. On December 12th, 2024, at 7 PM, we observed several fish deaths and would like to request that the relevant authorities investigate the cause and implement appropriate mitigation measures.

Question asked by a representative of Nam Yiem village, and responses by Xiengkhouang provincial department of natural resources and environment and vice district governor is as follows.

Thank you for the information. The project steering and compensation committee, which including provincial department of natural resources and environment, district leadership and authorities will immediately work together to carry the investigation and take an appropriate mitigation measure.

Q6. I would like to raise a concern regarding dust and proposed mitigation, for example, regular watering should be carried out by the project.

Question asked by a representative of Nam Khiao village, and responses by Xiengkhouang provincial department of natural resources and environment, vice district governor and Phongsubthavy Group Sole Co., Ltd is as follows.

The project is responsible for controlling dust, for example controlling speed of truck and watering the access roads where project-related transport occurs.

2.1.3 Free Prior and Informed Consent

Obtaining consent	No specific prior consent needs to be taken for the project as this is a voluntary project by the project proponent. There is no unresolved issue between the PP and the identified stakeholders.
Outcome of FPIC	As the proposed project activity is to install and operate 8.1 MW hydropower plant in existing land already acquired for the project, in line with Article 11 of EIA Decree of October 2022, the project has achieved IEE.

2.1.4 Grievance Redress Procedure

Development process	The grievance procedure was discussed during local stakeholder meeting. As, discussed the PP shall maintain a grievance register at site, wherein any stakeholder can provide feedback/inputs if any concerns. The PP has provided email IDs of concerned persons at site and for escalation if needed. Any concern raised will be initially taken up by team at site, however if no satisfactory solutions provided the stakeholder can escalate the same to General Manager, the concerns raised will be taken and resolved within agreed time frame as reasonable.
Grievance redress procedure	PP has provided contact details of site in-charge and project manager for raising any concern or feedback related to project activity. The project participant has established a record keeping system and if any feedback/complain received shall be recorded and remedy to same will be done with information to concern person/entity. If any concern/feedback received will be discussed and if require stakeholder will be invited again to discuss and take necessary action as may be needed.

2.1.5 Public Comments

Comments received	Actions taken
<i>To be added</i>	<i>To be added</i>

2.2 Risks to Stakeholders and the Environment

2.2.1 Management Experience

The promoters have long term experience in developing and operating hydro power projects. The company has a team of diverse experience.

The project activity is developed and operated by Nam Boak Hydropower Sole Co., Ltd, which is a subsidiary of Phongsubthavy Group Sole Co., Ltd has a proven track record for operation of similar projects in Host Country. They have a team that has experience in working on similar project area and are well-versed at their work.

2.2.2 Risk Assessment

	Risks identified	Mitigation or preventative measure(s) taken
Natural and human-induced risks to stakeholders' wellbeing	No risk identified	There were no natural or human induced risk identified due to proposed grouped project. As all the instances under grouped project are run-of -river project with no habilitation or displacement.
Risks to stakeholder participation	No risk identified	The stakeholders were provided with brief information related to project and agenda before stakeholder consultation process and there were no unresolved concern/conflict with any relevant stakeholders to the project. Further this project has no air pollution, no water pollution and no noise pollution. Hence, no potential risk identified which may impact stakeholder participation to the project.
Working conditions	No risk identified	The PP will provide health and safety training and wearing safety equipment's is mandatory at site to ensure no potential risk to worker. Further, regular health checkup is organized for workers periodically. Further, for worker and staffs at site, proper sanitation and hygiene facilities are maintained. Hence, no potential risk identified.
Safety of women and girls	Women and girls may face safety risks, including gender-based violence, harassment, and discrimination	Women and girls are of utmost importance to the PP. Measures will be for their protection as they are equally involved in the project activity. PP has Implemented gender-sensitive policies and practices that promote equal participation and empowerment of women and girls in

		decision-making processes related to the project. Provide gender-specific training and awareness programs on rights, safety, and reporting mechanisms to mitigate risks of gender-based violence and harassment.
Safety of minority and marginalized groups, including children	No risk identified	The project activity does not impact minority and marginalized groups including children. The project strictly complies with national legislation that explicitly forbids both forced labour and child labour.
Pollutants (air, noise, discharges to water, generation of waste, and release of hazardous materials and chemical pesticides and fertilizers)	No risk identified	The project does not produce any pollution in process of power generation. Further this project will have no air pollution, no water pollution and no noise pollution. The waste generation at project site is insignificant, further small quantity of cotton waste, waste oil and grease generated during maintenance is disposed through third party vendors. Hence, no potential risk identified.

2.3 Respect for Human Rights and Equity

2.3.1 Labor and Work

	Risks identified ³	Mitigation or preventative measure(s) taken
Discrimination	No risk identified	No discrimination has occurred as off now and the same will be ensured by the PP, that no such incident will happen in the future.
Sexual harassment	No risk identified	No sexual harassment has occurred as off now and the same will be ensured by the PP, that no such incident will happen in the future.
Equal pay for equal work	No risk identified	The human resource policy of the PP does not allow such discrimination and the pay is solely subject to skill and position and not based on gender.

³ The identified risks and commensurate mitigation or preventative measure(s) for forced labor, child labor, and human trafficking, must be inclusive of staff and contracted workers employed by third parties.

Gender equity in labor and work	No risk identified	The Project's positive developments has created jobs and skills for the local people. The Project Proponent has number of local women employments including management roles, thus increasing gender-equality and a gender-balanced workforce.
Forced labor	No risk identified	The Project Proponent will ensure that the project activity does not and will not use victims of human trafficking, forced labor, and child labor and policies will ensure the same. Immediate and strict measures will be taken in case PP comes across any such developments.
Child labor	No risk identified	The human resource policy of the PP does not allow child labor for any work.
Human trafficking	No risk identified	The human resource policy of the PP does not allow human trafficking.

2.3.2 Human Rights

Human rights abuses—such as child labor, forced labor, poor working conditions, gender inequality, and the violation of Indigenous land rights—are potential risks in many projects. As member state to the United Nations, Lao PDR has ratified several human rights instruments. The country has also ratified several Conventions of the International Labor Organization (ILO).

The human rights include the right to adequate food, the right to water and social & cultural rights. The project proponent will work to protect and promote the human rights of workers at project site. The project contributes in improving the environmental condition in the region of by generating emission free electricity using renewable resource i.e. hydro power. The project activity generated employment for skilled and unskilled workers due to project implementation and operation phase. The project activity supplies net electricity generated to grid, thereby results in emission reduction due to displacing equivalent electricity from fossil fuel based thermal power plants. As the land used is belongs to PP for respective project instance, hence no property or customary right of any stakeholders is impacted due to project activity.

Risks identified	Mitigation or preventative measure(s) taken
No risk identified	Not applicable

2.3.3 Indigenous Peoples and Cultural Heritage

Risks identified	Mitigation(s) or preventative measure taken
No risk identified	The project involves electricity generation using renewable resource i.e. hydro power. The project activity generated employment for skilled and unskilled workers due to project implementation and operation phase. As the land used belongs to PSG, no property or customary right of any stakeholders is impacted due to project activity. The PP commitment to uphold rights for indigenous peoples by ensuring transparency in project planning, execution and ultimately positive impact on nearby residents due to improved eco-system and employment opportunities

2.3.4 Property Rights

Risks identified	Mitigation or preventative measure(s) taken
No risk identified	The project instance involves electricity generation using renewable resource i.e. hydro power. The project activity will generate employment for skilled and unskilled workers due to project implementation and operation phase. As the land used belongs to PP, hence no property or customary right of any stakeholders is impacted due to project activity. The project does not impact rights of any stakeholder, hence no potential risk identified

2.3.5 Benefit Sharing

Process used to design the benefit sharing plan	Not applicable as the land used for the project activity belongs to PP. There, is no explicit benefit sharing arrangement with the local stakeholders of any kind neither is any such arrangement required as per local laws. However, PP plans to help nearby villagers through CSR activities in education, sports and other basic needs as per mutual agreement.
Summary of the benefit sharing plan	Not applicable.

Approval and dissemination of benefit sharing plan	Not applicable.
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2.4 Ecosystem Health

	Risks identified	Mitigation or preventative measure(s) taken
Impacts on biodiversity and ecosystems	No risk identified	The project instances under grouped VCS project are run-of-the-river projects. The construction for the project does not involve deep drilling. Hence, groundwater is not expected to be impacted during construction. Even during the operation the water, the flow downstream will take the natural course of the river after utilising the water without any impact on the groundwater. The Monitoring Plan accounts for measurement of the minimal water discharge and groundwater level at the site periodically during the operation phase. Further, the impact on the riverine and riparian habitat is very limited as the project uses trench-type diversion weir, which does not create any barrier for fish movement. The fish migrates during the summer season when there is plenty of water in the stream. Hence no potential risk identified.
Soil degradation and soil erosion	No risk identified	There is no negative impact due to project instances on soil. Moreover, PP plans plantation activity near project sites to prevent any potential soil erosion.
Water consumption and stress	No risk identified	There are no competing uses for water resources in the Project locations. Moreover, the project instances under grouped VCS project are a run-of-the-river projects. However, in line with the conservative practices of design, minimum discharge is maintained in river stream,

		which shall meet any other water requirements in the region for other social needs. The project instances will divert water stream for short distance and onwards releases the same to main river stream, however the same has no impact on water availability and does not create water stress.

2.4.1 Rare, Threatened, and Endangered Species

Is the project located in or adjacent to habitats for rare, threatened, or endangered species?

☐ Yes ☒ No

If yes, list such species and habitats in the table below and provide evidence that the project will not adversely impact these areas.

Species and habitat	There are no rare, threatened, or endangered species near project sites.
Areas needed for habitat connectivity	Not applicable.

2.4.2 Introduction of Species

The project activity is not a AFOLU project, hence not applicable.

2.4.3 Ecosystem Conversion

The project activity is not a AFOLU project, hence not applicable.

3 APPLICATION OF METHODOLOGY

3.1 Title and Reference of Methodology

Type (methodology, tool or module).	Reference ID, if applicable	Title	Version
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Methodology	AMS-I.D.	Grid connected renewable electricity generation	18.0
CDM-Tool	Tool 7	Tool to calculate the emission factor for an electricity system	7.0
CDM-Tool	Tool 21	Demonstration of additionality of small- scale project activity	13.1
CDM-Tool	Tool 27	Investment Analysis	14.0

3.2 Applicability of Methodology

The project activity involves generation of electricity by the means of renewable energy, hydro energy having installed capacity 8.1 MW. Hence, the project is a type I project and the project activity falls under the category of small-scale projects. The methodology chosen for the project activity and its applicability to the project activity is discussed below.

Methodology ID	Applicability condition	Justification of compliance
AMS-I.D.	1. This methodology comprises renewable energy generation units, such as photovoltaic, hydro, tidal/wave, wind, geothermal and renewable biomass a) Supplying electricity to a national or a regional grid b) Supplying electricity to an identified consumer facility via national/regional grid through a contractual arrangement such as wheeling.	The Project activity involves electricity generation using hydro power plant and supply of electricity to the national grid of Vietnam. Hence, criterion met.
AMS-I.D.	Illustration of respective situations under which each of the methodology (i.e. “AMS-I.D.: Grid connected renewable electricity generation”, “AMS-I.F.: Renewable electricity generation for captive use and mini-grid” and “AMS-I.A.: Electricity generation by the user) applies is included in the appendix.	As per Table No 2 of AMS – I. D. / Version 18, the AMS I.D is applicable to the project activity.
AMS-I.D.	This methodology is applicable to project activities that (a) Install a Greenfield plant; (b) Involve a capacity	The project involves installation of greenfield 8.1 MW hydro Power

	addition in (an) existing plant(s); (c) Involve a retrofit of (an) existing plant(s); (d) Involve a rehabilitation of (an) existing plant(s)/unit(s); or (e) Involve a replacement of (an) existing plant(s).	Plant at a site where no exiting power plant. Hence, criterion is met.
AMS-I.D.	Hydro power plants with reservoirs that satisfy at least one of the following conditions are eligible to apply this methodology: • The project activity is implemented in an existing reservoir with no change in the volume of reservoir; • The project activity is implemented in an existing reservoir, where the volume of reservoir is increased and the power density of the project activity, as per definitions given in the Project Emissions section, is greater than 4 W/m²; • The project activity results in new reservoirs and the power density of the power plant, as per definitions given in the Project Emissions section, is greater than 4 W/m²	The project activity is a hydro power project. Hence, the power density has been calculated by using formula of ACM0002 which is a CDM methodology. Since AMS-I.D does not include such a formula for it. The formula which is mentioned above is as follows: $PD = \frac{Cap_{PJ} - Cap_{BL}}{A_{PJ} - A_{BL}}$ where PD: Power density of the project activity (W/m²) CAP_{PJ}: Installed capacity of the hydro power plant after the implementation of the project activity (W) CAP_{BL}: Installed capacity of the hydro power plant before the implementation of the project activity (W). For new hydro power plants, this value is zero A_{PJ}: Area of the single or multiple reservoirs measured in the surface of the water, after the implementation of the project activity, when the reservoir is full (m²) A_{BL}: Area of the single or multiple reservoirs measured in the surface of the water, before the implementation of the project activity, when the reservoir is full (m²). For new reservoirs, this value is zero. Thus, PD= (8,100,000 -0) / (7,852-0) = 1,031.58 W/m² Where installed capacity is 8.1 MW and the area of reservoir is 0.007

		Km ² , and the power density is 1,031.58 W/m ² , so that the power density is greater than 4 W/m ² . For new hydro power projects, CAP _{BL} and A _{BL} is zero.
AMS-I.D.	If the new unit has both renewable and non- renewable components (e.g. a wind/diesel unit), the eligibility limit of 15MW for a small-scale CDM project activity applies only to the renewable component. If the new unit co fires fossil fuel ⁴ , the capacity of the entire unit shall not exceed the limit of 15MW.	The project activity is only 8.1 MW hydro energy based renewable electricity generation project. It does not include any non-renewable unit and co-firing system
AMS-I.D.	Combined heat and power (co-generation) systems are not eligible under this category.	The project activity does not involve combined heat and power generation system as it involves hydro energy-based power generation.
AMS-I.D.	In the case of project activities that involve the addition of renewable energy generation units at an existing renewable power generation facility, the added capacity of the units added by the project should be lower than 15 MW and should be physically distinct from the existing units.	The project activity is a greenfield project. Hence this criterion is not applicable.
AMS-I.D.	In the case of retrofit or replacement, to qualify as a small-scale project, the total output of the retrofitted or replacement unit shall not exceed the limit of 15 MW.	The project activity is not the retrofitting or replacement of an existing facility for renewable energy generation. Hence this criterion in not applicable.
AMS-I.D.	In the case of landfill gas, waste gas, wastewater treatment and agro-industries projects, recovered methane emissions are eligible under a relevant Type III category. If the recovered methane is used for electricity generation for supply to a grid then the baseline for the electricity component shall be in accordance with procedure prescribed under this methodology. If the recovered methane is used for heat generation or cogeneration other	The proposed project activity is a hydroelectric power project, hence criterion not applicable.

	applicable Type-I methodologies such as “AMS-I.C.: Thermal energy production with or without electricity” shall be explored.	
AMS-I.D.	In case biomass is sourced from dedicated plantations, the applicability criteria in the tool “Project emissions from cultivation of biomass” shall apply.	The proposed project activity is a hydroelectric power project, hence criterion not applicable.
Tool 7	This tool may be applied to estimate the OM, BM and/or CM when calculating baseline emissions for a project activity that substitutes grid electricity that is where a project activity supplies electricity to a grid or a project activity that results in savings of electricity that would have been provided by the grid (e.g. demand-side energy efficiency projects).	This condition is applicable. OM, BM and CM are estimated using the tool under section 4.1 for calculating baseline emissions. Therefore, the tool is applicable.
	Under this tool, the emission factor for the project electricity system can be calculated either for grid power plants only or, as an option, can include off-grid power plants. In the latter case, the conditions specified in “Appendix 2: Procedures related to offgrid power generation” should be met. Namely, the total capacity of off-grid power plants (in MW) should be at least 10 per cent of the total capacity of grid power plants in the electricity system; or the total electricity generation by off-grid power plants (in MWh) should be at least 10 per cent of the total electricity generation by grid power plants in the electricity system; and that factors which negatively affect the reliability and stability of the grid are primarily due to constraints in generation and not to other aspects such as transmission capacity.	Since the project activity is grid connected, this condition is applicable to the project activity and the emission factor has been calculated accordingly. Therefore, the tool is applicable.
Tool 7	In case of CDM projects the tool is not applicable if the project electricity system is located partially or totally in an Annex I country	The project activity is located in Lao PDR and generated electricity supplied to Vietnam grid, both are a non-Annex I country.

		Therefore, this tool is applicable for the project activity.
Tool 7	Under this tool, the value applied to the CO ₂ emission factor of biofuels is zero.	The project does not involve biofuels usage. Therefore, this criterion is not applicable for the project activity.
Tool 21	The use of the methodological tool “Demonstration of additionality of small-scale project activities” is not mandatory for project owner when proposing new methodologies. Project Owner and coordinating/managing entities may propose alternative methods to demonstrate additionality for consideration by the Executive Board.	The proposed project is a small-scale project and the additionally tool is included in the approved methodology. Therefore, the tool is applicable.
Tool 21	Project Owner and coordinating/managing entities may also apply “TOOL19: Demonstration of additionality of microscale project activities” as applicable.	The project is not a microscale project. Therefore, this criterion is not applicable for the project activity.
Tool 27	This methodological tool is applicable to project activities that apply the methodological tool “Tool for the demonstration and assessment of additionality”, the methodological tool “Combined tool to identify the baseline scenario and demonstrate additionality”, the guidelines “Non-binding best practice examples to demonstrate additionality for SSC project activities”, or baseline and monitoring methodologies that use the investment analysis for the demonstration of additionality and/or the identification of the baseline scenario.	Project activity applies “Tool for the demonstration and assessment of additionality”. Therefore, the tool is applicable.
	In case the applied approved baseline and monitoring methodology contains requirements for the investment analysis that are different from those described in this methodological tool, the requirements contained in the methodology shall prevail	The project activity is a grid connected hydro power project. The project refers to the “Tool for the demonstration and assessment of additionality”. Therefore, the tool is applicable.

3.3 Project Boundary

As per Para 18 of applied baseline and monitoring methodology AMS I.D, Version-18, the spatial extent of the project boundary includes the project power plant and all power plants connected physically to the electricity system that the VCS project power plant is connected to. This includes the turbine, generator and sub-stations. The project will supply electricity to national grid of Vietnam using direct transmission line from the project, hence the same is adopted as the project boundary.

The proposed project activity will supply generated power to the regional grid. Therefore, all the power plants contributing electricity to the Vietnam Grid have been considered in the project boundary for the purpose of baseline estimation. The project activity targets reduction of CO₂e as main GHG greenhouse gas in baseline, there are no GHG emission associated with project activity.

The table below provides an overview of the emissions sources included or excluded from the project boundary for determination of baseline and project emissions

Source		Gas	Included?	Justification/Explanation
Baseline	Electricity generation	CO ₂	Yes	Main emission source
		CH ₄	No	Excluded for simplification
		N ₂ O	No	Excluded for simplification
Project	For geothermal power plants, fugitive emissions of CH ₄ and CO ₂ from non-condensable gases contained in geothermal steam.	CO ₂	No	Not Applicable, as project activity is a hydro power plant.
		CH ₄	No	
		N ₂ O	No	
	CO ₂ emissions from combustion of fossil fuels for electricity generation in solar thermal power plants and geothermal power plants.	CO ₂	Yes	The project has diesel based DG set and any diesel consumption will be monitored ex-post.
		CH ₄	No	
		N ₂ O	No	
Leakage	For hydro power plants, emissions of CH ₄ from the reservoir.	CO ₂	No	Minor emission source. Excluded for simplification
		CH ₄	Yes	Main emission source
		N ₂ O	No	Minor emission source. Excluded for simplification

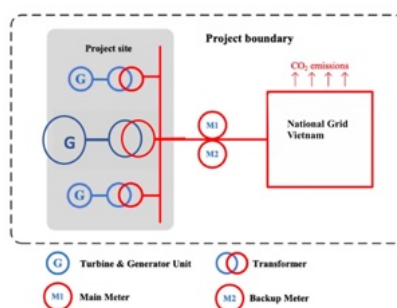


Fig: Project boundary

3.4 Baseline Scenario

The proposed project activity involves installation and operation of 8.1MW greenfield hydro power project. The generated power will be supplied to Vietnam Grid, which otherwise would have been generated from grid connected power plants, which possesses a mix of generation types with fossil fuel fired power plants.

As per para 19 of AMS-I.D. (Version 18) “The baseline scenario is that the electricity delivered to the grid by the project activity would have otherwise been generated by the operation of grid-connected power plants and by the addition of new generation sources into the grid.” In the absence of the project activity same amount electricity would have been generated from grid, in which the electricity is generated by the fossil fuel intensive power plant. Thus, baseline is in line with para 19 of AMS-I.D. (Version 18).

Para 22 of AMS-I.D. (Version 18) calculates baseline emissions as:

$$BE_y = EG_{PJ,y} \times EF_{grid,y} \dots\dots\dots(A)$$

Where,

BE_y = Baseline Emissions in year y; t CO₂

$EG_{PJ,y}$ = Quantity of net electricity generation that is produced and fed into the grid as a result of the implementation of the CDM project activity in year y (MWh)

$EF_{grid,y}$ = Combined margin CO₂ emission factor for grid connected power generation in year y calculated using the latest version of the “Tool to calculate the emission factor for an electricity system” (t CO₂/MWh)

Accordingly, the emission factor of the grid will be used to estimate emission reductions. As per para 23 of AMS-I.D. (Version 18), PP has chosen option (a) and used the combined margin (CM) approach to calculate emission factor, as official data is available for operating margin (OM) and build margin (BM) values, whereas no such data exists in the public domain to support choice of option (b).

Hence,

$$EF_{grid,y} = EF_{grid,CM,y} \dots(B)$$

Data Used-

Parameter	Description	Source
EF _{OM,Y}	Operating margin CO ₂ emission factor for the project electricity system in year y	Calculated as per “Tool to calculate the emission factor for an electricity system (Version 07.0.0)”
EF _{BM,Y}	Build margin CO ₂ emission factor for the project electricity system in year y	
EF _{CM,Y}	Combined margin CO ₂ emission factor for the project electricity system in year y	
EG _{PJ,Y}	Quantity of net electricity supplied by the candidate project activity to the grid in year y 28.64 GWh	As per FSR prepared by third party engineering consultant
EF _{OM,Y}	0.9239tCO ₂ /MWh	Please refer section 4.1 of PD and ER sheet
EF _{BM,Y}	0.5202tCO ₂ /MWh	Please refer section 4.1 of PD and ER sheet
EF _{CM,Y}	0.72205tCO ₂ /MWh	Please refer section 4.1 of PD and ER sheet
Electricity generation	28.64 GWh/year	Third party report in line with para 3(a) of EB48 Annex-11

3.5 Additionality

3.5.1 Regulatory Surplus

Is the project located in an UNFCCC Annex 1 or Non-Annex 1 country?

☐ Annex 1 country ☒ Non-Annex 1 country

Are the project activities mandated by any law, statute, or other regulatory framework?

☐ Yes ☒ No

If the project is located inside a Non-Annex 1 country and the project activities are mandated by a law, statute, or other regulatory framework, are such laws, statutes, or regulatory frameworks systematically enforced?

☐ Yes ☒ No

The project is not enforced by law. The project passes the legal requirement test since there are no enforced laws, statutes, regulations, court orders, environmental-mitigation agreements, permitting conditions of other legally-binding mandates requiring its implementation. Since voluntary commitments/agreements within a sector or by an entity do not constitute the legal requirement.

Mandatory laws and regulations are present:

Law on Electricity & Decree (1997)
 Law on Environmental Protection (1999)
 Electricity Law (2017)
 Law on Forestry (revised) (2021)

The above referenced regulation does not mandate implementation of hydro power project.

3.5.2 Additionality Methods

According to Demonstration of additionality of small-scale project activities (Version 14.0), Project participants shall provide an explanation to show that the project activity would not have occurred anyway due to at least one of the following barriers:

- a) Investment barrier: a financially more viable alternative to the project activity would have led to higher emissions;
- b) Technological barrier: a less technologically advanced alternative to the project activity involves lower risks due to the performance uncertainty or low market share of the new technology adopted for the project activity and so would have led to higher emissions;
- c) Barrier due to prevailing practice: prevailing practice or existing regulatory or policy requirements would have led to implementation of a technology with higher emissions;
- d) Other barriers: without the project activity, for another specific reason identified by the project participant, such as institutional barriers or limited information, managerial resources, organizational capacity, financial resources, or capacity to absorb new technologies, emissions would have been higher.

Chronology of events:

Activity	Date	Source
Feasibility Study Report (FSR)	July 2020	FSR
Investment Decision Date	09 December 2020	Investment decision letter
EIA/IEE	20/05/2021	IEE certificate
Purchase Order/EPC	28 June 2022	EPC Agreement
Power Purchase Agreement	28 June 2021	PPA
Commissioning	01 October 2025	Expected date of Commissioning

The additionality assessment is based on the proposition that the project faces an investment barrier would prevents its implementation. As a small hydropower project located in poor mountainous area, the project faces many implementation complexities, which make it hardly financial attractive. The investment barrier represents the most prohibitive factor in implementing the project.

The project activity reduces anthropogenic emissions of greenhouse gases that would have occurred in absence of the project activity. As per the 6 (c) decision 17/CP.7⁴ Para 43, a VCS project activity is additional if anthropogenic emissions of greenhouse gases by sources are reduced below those that would have occurred in the absence of the registered VCS project activity.

As per the investment analysis, the project participant is required to determine whether the project activity is economically or financially less attractive than other alternatives without the revenue from the sale of Verified Carbon Units (VCUs). The project activity will sell the generated electricity to grid and has financial implications other than those related to VCUs.

Moreover, as the project activity will supply generated electricity to grid, hence the project can be developed by any entity and PP is not having any requirement to invest in the project. In line with para 18, In the cases of projects which could be developed by an entity other than the project participant the benchmark should be based on parameters that are standard in the market.

Accordingly, PP has adopted the benchmark analysis wherein the financial indicator(s) of the project activity is compared with standard benchmark value as in case baseline scenario is limited to very small investment which in not significant in comparison to project activity.

Selection of financial indicator:

In order to analyse the financial viability of the project activity, the prime financial indicator that has been used is the equity IRR of the project activity. The equity IRR is one of the most commonly used tools to assess the feasibility and viability of the projects. Since the project developer is demonstrating the financial unattractiveness of the project, equity IRR is appropriate as it is often used by the project developers to make a decision on investing in the project. Hence, equity IRR is considered the financial indicator for demonstrating the additionality of the project.

Selection of benchmark:

As per para 20 of tool Investment Analysis, If the benchmark is based on parameters that are standard in the market, the cost of equity should be determined either by: (a) selecting the values provided in Appendix; or by (b) calculating the cost of equity using CAPM.

PP has considered the value of 20.98% provided as the default Return on Equity (ROE) value in Appendix-Default values for cost of equity (return on equity) of the Investment Analysis tool.

⁴ <http://unfccc.int/resource/docs/cop7/13a02.pdf#page=36>

Further as per tool PP has chosen option a) selecting the values provided in Appendix; Default values for cost of equity (return on equity), Project participant has considered the benchmark based on equity indices. Cost of equity has been estimated based on the values provided in Appendix-Default values for cost of equity (return on equity). As per the values, Cost of Equity can be estimated as below –

$$\text{Return on equity}_{\text{Nominal}} = (1 + \text{Return on equity}_{\text{Real}}) * (1 + \text{Inflation rate}_{\text{Host country}}) - 1$$

Apart from the default values of ROE provided in the Appendix, the para 16 of tool is also used to finally arrive at the value of the benchmark. As per the same, "In situations where an investment analysis is carried out in nominal terms and the available IRR benchmarks are in real terms, project participants shall convert the real term values of benchmarks to nominal values by adding the inflation rate. The inflation rate shall be obtained from the inflation forecast of the central bank of the host country for the duration of the crediting period. If this information is not available, the target inflation rate of the central bank shall be used. If this information is also not available, then the average forecasted inflation rate for the host country published by the IMF (International Monetary Fund World Economic Outlook) or the World Bank for the next five years after the start of the project activity shall be used".

As long term forecast on inflation by host country was not available, 5 years inflation forecast by World bank is used. World Bank forecasted values for the next 5 years for inflation has been used to adjust the default value of ROE, which is given in real terms as per below for the proposed project activity.

Long term inflation for 5 years forecast applicable to project activity i.e. 4.9%⁵ is applicable under proposed project activity as the same was available to PP at that time of investment decision date i.e. 09-December-2020.

Therefore,

$$\text{Return on equity}_{\text{Nominal}} = (1 + 20.98\%) * (1 + 4.9\%) - 1 = 26.91\%$$

Hence, the return on equity 26.91% has been considered as benchmark value.

Basic parameters for calculation of financial indicators

The financial index from FSR was used for equity IRR analysis. FSR was completed by third party in July 2020 which is earlier than the project start date.

⁵ <https://data.worldbank.org/indicator/FP.CPI.TOTL.ZG?locations=LA>

<https://documents1.worldbank.org/curated/en/604471565799670466/pdf/Lao-PDR-Economic-Monitor-Maintaining-Economic-Stability.pdf>

<https://pubdocs.worldbank.org/en/306131610167443727/Lao-PDR-Economic-Monitor-January-2021-Supporting-Economic-Recovery-Thematic-section-Livelihoods-in-the-Time-of-COVID-19.pdf>

The equity IRR for the project activity is calculated considering normal lifetime of turbines/generator of 25 years, which is appropriate as onwards the significant repair or maintenance will be required.

From the above analysis, the FSR value is more conservative, which is used for the VCS Project financial analysis and Critical value analysis, the main assumptions for the investment analysis are shown in Table below.

Parameter	Value	Source
Installed Capacity of each turbine in kW	2,700	Feasibility Study Report
Number of Turbine	3	Feasibility Study Report
Project Capacity in MW	8.1	Feasibility Study Report
Date of commissioning	01/10/2025	Expected date of Commissioning
Capital Cost		
Project Cost (USD Millions) Including IDC	14,806.71	Feasibility Study Report
Construction period (months)	36	Feasibility Study Report
Loan Amount (USD Millions)	9,680	Feasibility Study Report
Equity Amount (USD Millions)	4,497	Feasibility Study Report
O&M and other recurring expenses		
Interest rate	8%	Feasibility Study Report
Water Royalty to be paid to Govt. from 1 st year	5%	Feasibility Study Report
Operation & Maintenance Cost		
O&M@ capital cost	1.5%	Feasibility Study Report
Generation		
Gross Electricity Generation per year (GWh)	28.84	Feasibility Study Report
Auxiliary consumption	1%	Feasibility Study Report

Net Electricity Generation per year (GWh)	28.552	Calculated
Tariff electricity USD/kWh	0.0695	Feasibility Study Report/PPA
Book Depreciation and Residual value		
Book depreciation percentage of total project cost	4%	Feasibility Study Report
Income Tax		
Income tax from commissioning	24%	Feasibility Study Report
Salvage value	10%	Considered as conservative
Lifetime of the project activity (years)	25	Feasibility Study Report,
Investment analysis period (years)	25	

Based on above the equity IRR comes out to be 17.32% which is well below benchmark value of 26.91%.

Sub-step 2d. Sensitivity analysis

The sensitivity analysis shows whether the conclusion regarding financial attractiveness is robust to reasonable variations in the critical assumptions. For the project, the most important parameters impacting the equity IRR are:

- Fixed assets investment
- Annual O&M cost
- Electricity tariff
- Power supplied to the grid

In case of the $\pm 10\%$ variation range of the four parameters, the fluctuations of the equity IRR are showing below:

SENSITIVITY ANALYSIS				
Equity IRR	17.32%			
Benchmark	26.91%			
Parameter	-10%	Base case	10%	Breaching Value

Project Cost	21.71%	17.32%	14.59%	-16.73%
O&M	17.57%	17.32%	17.07%	-435.40%
Electricity generation	13.09%	17.32%	21.76%	22.12%
Electricity Tariff	15.09%	17.32%	19.49%	49.12%

Project cost:

The sensitivity analysis reveals that project cost decrease by 16.73% will breach the benchmark, however the actual cost incurred is expected to be higher than used for investment analysis, hence it's not a possible scenario for the project activity.

O&M:

The decrease in O&M cost does not have significant impact on IRR, even 100% removal of O&M cost will not be breaching benchmark value. Hence its hypothetical case as cost cannot be less than 100% than estimated.

Electricity Tariff:

The sensitivity analysis reveals that an increase in electricity tariff by 49.12% will breach the benchmark, however the PP has already long term PPA in place with same tariff structure, hence it's not a possible scenario for the project activity.

Electricity Generation:

The sensitivity analysis reveals that an increase in power generation by 22.12% will breach the benchmark, however the power generation potential is established based on long term hydrology study of catchment area and precipitation pattern and increase of more than double of estimated PLF is not a possibility, hence it's not a possible scenario for the project activity.

Based on the relationship shown above, we can find out that the equity IRR that will decline accompany with the rise of the fixed assets investment and the annual O&M cost; and the equity IRR will rise accompany with the rise of the electricity tariff and the electricity supply. We can conclude from the above analysis that, even if $\pm 10\%$ variation range of the four parameters, the equity IRR of the project still can't surpass the benchmark. However, the revenue from the VCUs will improve the funding gap for the project.

In conclusion, without the consideration of the revenue from VCUs, the conclusion of the proposed VCS project activities lacks of commercial attraction is evidenced, so the specific project is in shortage of commercial attraction. Hence project is additional.

3.6 Methodology Deviations

The project activity has not taken any deviation from the applied approved methodology AMS-I.D.: Grid connected renewable electricity generation, Version 18.0.:

4 QUANTIFICATION OF ESTIMATED GHG EMISSION REDUCTIONS AND REMOVALS

4.1 Baseline Emissions

As per para 22 of AMS-I.D. (Version 18), baseline emissions are calculated as follows:

$$BE_y = EGP_{J,y} \times EF_{grid,y}$$

Where:

BE_y = Baseline Emissions in year y; t CO₂

$EGP_{J,y}$ = Quantity of net electricity generation that is produced and fed into the grid as a result of the implementation of the CDM project activity in year y (MWh)

$EF_{grid,y}$ = Combined margin CO₂ emission factor for grid connected power generation in year y calculated using the latest version of the “Tool to calculate the emission factor for an electricity system” (t CO₂/MWh)

Calculation of $EGP_{J,y}$

As proposed project activity is a new distinct unit 10MW hydropower plant (Greenfield) project commissioned next to existing 76MW hydropower plant and does not impact the electricity generation potential of the existing project. Further, the project activity is not a rehabilitation or retrofit of existing unit, hence in accordance with para 26 of applied methodology

$$EGP_{J,y} = EGP_{J, facility,y}$$

Where,

$EGP_{J, facility,y}$ = Quantity of net electricity generation supplied by the project plant/unit to the grid in year y (MWh)

Calculation of BE_y

Calculation of baseline emissions i.e. BE_y , requires calculation of grid emission factor ($EF_{grid,y}$), which is being presented below.

The project activity is located in host country Lao PDR, however it supplied net generated electricity to Vietnam National grid through direct transmission line without using grid transmission line of the Lao PDR, hence identified electricity system in project boundary is Vietnam National grid.

As per para 23 of the applied methodology, the emission factor can be calculated in a transparent and conservative manner as follows:

Emission factor of a grid has been calculated as per the procedures provided in AMS-I.D;

For a mini-grid system other than described in paragraph 18 above, the baseline emission factor shall be determined as per the weighted average emissions for the current generation mix following the procedure provided in AMS-I. D;

Emission factor for captive electricity generation shall be calculated as per the procedures described in the latest version of the "Tool to calculate baseline, project and/or leakage emissions from electricity consumption"

The emission factor of the grid will be used to estimate emission reductions. As per para 23 of AMS-I.D. (Version 18), PP has chosen option (a) and used the combined margin (CM) approach to calculate emission factor, as official data is available for operating margin (OM) and build margin (BM) values, whereas no such data exists in the public domain to support choice of option (b). Hence,

$$EF_{CM,grid,y} = EF_{CM,y} \quad \dots(B)$$

The PP has chosen option a i.e. combined margin (CM) consisting of combination OM and BM. Tool to calculate the emission factor for an electricity system (Version 07.0.0), has been used to determine the CO2 emission factor for displacement of electricity generated by power plants in an electricity system, by calculating the combined margin emission factor (CM) of that electricity system.

As per the "Tool to calculate the emission factor for an electricity system" Version 07.0, EB 100, Annex 4, the following steps have been followed.

As per the methodology combined margin grid emission factor has been calculated as per the "Tool to calculate the grid emission factor for an Electricity System" version 07.

Most updated national grid data of Vietnam: The data on national grid emission factors was published by Department of Climate Change - Ministry of Natural Resources and Environment, Research and develop emission factor (EF) of Vietnam's electricity grid in 2021 (attached with CV 1278/BDKH-TTBVTOD)" (ref. [http://dcc.gov.vn/van-ban-phap-luat/1102/Nghien-cuu-xay-dung-he-so-phat-thai-\(EF\)-cua-luoi-dien-Viet-Nam-nam-2021-\(kèm-CV-1278/BDKH-TTBVTOD\).html](http://dcc.gov.vn/van-ban-phap-luat/1102/Nghien-cuu-xay-dung-he-so-phat-thai-(EF)-cua-luoi-dien-Viet-Nam-nam-2021-(kèm-CV-1278/BDKH-TTBVTOD).html)) has been used for the calculation of emission factor.

As per the "Tool to calculate the emission factor for an electricity system" Version 07.0, EB 100, Annex 4, the following steps have been followed.

Step 1: Identify the relevant electricity systems;

Step 2: Choose whether to include off-grid power plants in the project electricity system (optional);

Step 3: Select a method to determine the operating margin (OM);

Step 4: Calculate the operating margin emission factor according to the selected method;

Step 5: Calculate the build margin (BM) emission factor;

Step 6: Calculate the combined margin (CM) emission factor.

Step 1: Identify the relevant electricity systems

For the purpose of determining the electricity emission factor, the project electricity system is defined as the electricity transmission system of Vietnam which is a single system connected by transmission lines throughout the country and owned by the Vietnam Electricity Corporation (EVN). Electricity imports from a connected electricity system are included and as per Tool to calculate emission factor of an electricity system, for the purpose of determining the operating margin emission factor, 0 tCO₂/MWh is applied.

Step 2: Choose whether to include off-grid power plants in the project electricity system (optional)

Only grid connected power plants are included in the calculation, as per Option I of the “Tool to calculate the emission factor for an electricity system” version 07.0.0

Step 3: Select a method to determine the operating margin (OM)

The calculation of the operating margin emission factor ($EF_{grid,OM,y}$) is based on one of the following methods, which are described under

Step 4: (a) Simple OM; or

(b) Simple adjusted OM; or

(c) Dispatch data analysis OM; or

(d) Average OM.

The simple OM method (Option a) is used for this study as the low-cost/must-run resources (LCMR) constitute less than 50% of total electricity production. The share of low-cost/must run power plants in the last five years are around 38.03% as shown in the below table.

Table: Percentage of electricity output from LCMR sources

Year	2017	2018	2019	2020	2021	Total
Type of Power Plants						
Hydropower	71,056,945	69,485,682	54,411,106	59,387,446	69,606,845	323,948,025
Bagasse	78,000	456,400	280,996	331,319	347,560	1,494,275
Wind Power	-	-	721,189	946,157	3,243,227	4,910,574

Solar Power	-	-	4,833,674	9,684,525	15,141,520	29,659,719
Imported electricity	2,361,000	3,124,000	3,316,000	3,067,000	1,401,463	13,269,463
Total Power Output	169,942,517	188,063,484	207,214,694	207,692,796	208,561,267	
LCMR ratio						38.03%

Data Source: Vietnamese grid emission factor in 2021, Ministry of Natural Resources and Environment, Department of Climate Change⁶

As per tool to calculate emission factor for an electricity system (Version 07), The simple OM method (option a) can only be used if low-cost/must-run resources constitute less than 50% of total grid generation in: 1) average of the five most recent years, or 2) based on long-term averages for hydroelectricity production. Since the low cost/must run resources constitute less than 50% of total grid generation as seen from the average of five most recent years, the Simple OM method can be used to calculate the Operating Margin Emission factor. PP has chosen ex ante option, thus, no monitoring and recalculation of the emissions factor during the crediting period is required.

PP has considered a data vintage of 3-year generation-weighted average, based on the most recent data available at the time of submission of the VCS PD for listing at VERRA.

Step 4: Calculate the operating margin emission factor ($EF_{grid,OMSimple,y}$) according to the selected method

The simple OM emission factor is calculated as the generation-weighted average CO₂ emissions per unit net electricity generation (tCO₂/MWh) of all generating power plants serving the system, not including low-cost / must-run power plants / units.

The simple OM may be calculated:

Option A: Based on the net electricity generation and a CO₂ emission factor of each power unit;
Or

Option B: Based on the total net electricity generation of all power plants serving the system and the fuel types and total fuel consumption of the project electricity system.

Department of Climate Change - Ministry of Natural Resources and Environment, "Research and develop emission factor (EF) of Vietnam's electricity grid in 2021 (attached with CV 1278/BDKH-TTBVTOD)" published dated 21/12/2022. This provides information about the Combined Margin Emission Factor of Vietnam national grid. The Combined Margin is calculated ex ante using the guidelines provided by the UNFCCC in the "Tool to calculate the emission factor for an electricity

⁶ [http://dcc.gov.vn/van-ban-phap-luat/1102/Nghien-cuu-xay-dung-he-so-phat-thai-\(EF\)-cua-luoi-dien-Viet-Nam-nam-2021-\(kèm-CV-1278/BDKH-TTBVTOD\).html](http://dcc.gov.vn/van-ban-phap-luat/1102/Nghien-cuu-xay-dung-he-so-phat-thai-(EF)-cua-luoi-dien-Viet-Nam-nam-2021-(kèm-CV-1278/BDKH-TTBVTOD).html)

system, Version 07". We have, therefore, used the Combined Margin data published in the Grid Emission Factor for GHG Reduction Project/Activity, for calculating the Baseline Emission Factor.

As per "Tool to calculate the emission factor for an electricity system", Option B ("Calculation based on total fuel consumption and electricity generation of the system") is used to calculate simple OM emission factor. Where Option B is used, the simple OM emission factor is calculated based on the net electricity supplied to the grid by all power plants serving the system, not including low cost/must run power plants/units, and based on the fuel type(s) and total fuel consumption of the project electricity system, as follows:

$$EF_{\text{grid,OM, ave,y}} = \sum (FC_{i,y} \times NCV_{i,y} \times EF_{\text{CO}_2,i,y}) / EG_y$$

Where:

$EF_{\text{grid,OM, ave,y}}$ = Average operating margin CO₂ emission factor in year y (tCO₂/MWh);

$FC_{i,y}$ = Amount of fossil fuel type i consumed in the proposed CDM project electricity system in year y (mass or volume unit);

$NCV_{i,y}$ = Net calorific value (energy content) of fossil fuel type i in year y (GJ /mass or volume unit);

$EF_{\text{CO}_2,i,y}$ = CO₂ emission factor of fossil fuel type i in year y (tCO₂/GJ);

EG_y = Net electricity generated and delivered to the grid by all power sources serving the system, including low-cost/must-run power plants/units, in year y (MWh);

i = All fossil fuel types combusted in power sources in project electricity system in year y;

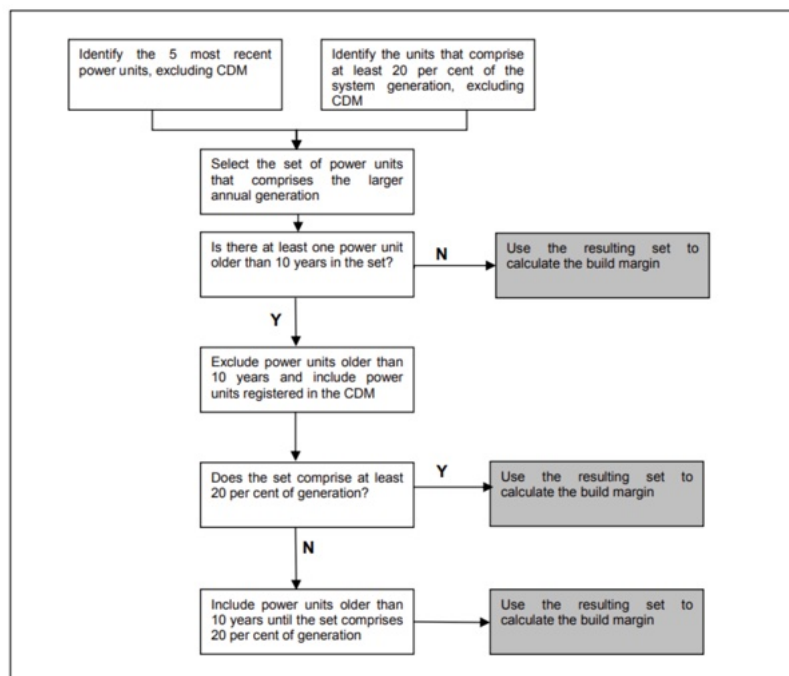
y = The data available in the most recent 3 years;

According to the Tool to calculate the emission factor for an electricity system, electricity imports from the connected electricity systems $EG_{\text{import,y}}$ are included in the EG_y .

As per the Vietnamese grid emission factor in 2021 weighted average operating margin is as below:

$$EF_{\text{grid,OM,y}} = 0.9239 \text{ tCO}_2/\text{MWh}$$

STEP 5: Calculate the build margin emission factor ($EF_{\text{BM,y}}$) The project participants have chosen Option I, i.e. fixing build margin emission factor ex ante based on the most recent information available on units already built for sample group m at the time of PD submission to the VERRA for listing. The build margin emissions factor is the generation-weighted average emission factor (tCO₂/MWh) of a sample group of power units, during the most recent year y for which power generation data is available. The Sample group of power units m used to calculate the build margin should be determined via the procedure summarized in the diagram of the Tool



Following this procedure, the list of plants/units selected to calculate the marginal emission factor built in 2021. The total power output of the plants commissioned in 2021 is accounting for 20.24% of the total electricity output of Vietnam's electricity grid in 2021. Using the equation given in the step 5 for the BM calculation, the Built margin is calculated for the year 2021 is as below:

$$EF_{grid,BM,y} = 0.5202 \text{ tCO}_2/\text{MWh}$$

STEP 6: Calculate the combined margin (CM) emissions factor

The combined margin is the weighted average of the simple operating Margin and the build margin.

$$EF_{grid,CM,y} = EF_{grid,OM,y} \times W_{OM} + EF_{grid,BM,y} \times W_{BM}$$

Where,

$EF_{grid,BM,y}$ = Build margin CO₂ emission factor in year y (tCO₂/MWh)

$EF_{grid,OM,y}$ = Operating margin CO₂ emission factor in year y (tCO₂/MWh)

W_{OM} = Weighting of operating margin emission factor (per cent)

W_{BM} = Weighting of build margin emission factor (per cent)

In particular, the Tool to calculate the emission factor for an electricity system, Version 07.0.0, EB 100, Annex 4, allows to weigh the operating margin and Build margin at 50% and 50%, respectively for hydro power projects. As the project activity is hydro power project the above weight is applied to calculated combined margin emission factor.

The baseline emission factor EF_y is calculated as the weighted average of the Operating Margin emission factor ($EF_{OM,y}$) and the Build Margin emission factor ($EF_{BM,y}$) as below:

$$EF_{grid,CM,y} = 0.9239 \times 0.5 + 0.5202 \times 0.5$$

$$EF_{grid,CM,y} = 0.7221 \text{ tCO}_2/\text{MWh}$$

The grid emission factor established above is fixed ex-ante for first crediting period.

Electronic spreadsheet showing calculation of all these parameters is being submitted separately and the final values are presented below:

Parameter	Value	Units
Operating Margin : $EF_{OM,y}$	0.9239	tCO _{2e} /MWh
Build Margin : $EF_{BM,y}$	0.5202	
Combined Margin : $EF_{grid,CM,y}$	0.7221	

4.2 Project Emissions

As per the approved consolidated Methodology AMS I.D (Version 18.0, EB 81): “For most renewable energy power generation project activities, $PE_y = 0$.

As per applied methodology only emission associated with the fossil fuel combustion, emission from operation of geo-thermal power plants due to release of non-condensable gases, emission from water reservoir of Hydro should be accounted for the project emission. Since the project activity is a run of river hydro power project,

CO₂ emissions from on-site consumption of fossil fuels due to the project activity shall be calculated using the Methodological tool “Tool to calculate project or leakage CO₂ emissions from fossil fuel combustion” (Version 3.0). The project activity has installed of one DG set to meet the emergency requirements of power house. Emissions due to use of diesel in DG set will be accounted as project emissions based on the following derived equation based on tool.

CO₂ emissions from fossil fuel combustion in process j are calculated based on the quantity of fuels combusted and the CO₂ emission coefficient of those fuels, as follows:

$$PE_{FC,j,y} = \sum FC_{i,j,y} \times CO_{EFi,y}$$

Where: $PE_{FC,j,y}$ = Are the CO₂ emissions from fossil fuel combustion in process j during the year y (tCO₂/yr)

$FC_{i,j,y}$ = Is the quantity of fuel type i combusted in process j during the year y (mass or volume unit/yr)

$CO_{EFi,y}$ = Is the CO₂ emission coefficient of fuel type i in year y (tCO₂/mass or volume unit)

i = Are the fuel types combusted in process j during the year y

Since the chemical composition of fossil fuel is not available with PP, the option B of tool is followed for CO₂ emission coefficient $CO_{EFi,y}$

Option B: The CO₂ emission coefficient $CO_{EFi,y}$ is calculated based on net calorific value and CO₂ emission factor of the fuel type i, as follows:

$$CO_{EFi,y} = NCV_{i,y} \times EF_{CO2,i,y}$$

Where: $CO_{EFi,y}$ = Is the CO₂ emission coefficient of fuel type i in year y (tCO₂/mass or volume unit)

$NCV_{i,y}$ = Is the weighted average net calorific value of the fuel type i in year y (GJ/mass or volume unit)

$EF_{CO2,i,y}$ = Is the weighted average CO₂ emission factor of fuel type i in year y (tCO₂/GJ)

i = Are the fuel types combusted in process j during the year y

As per tool, IPCC default values at the upper limit of the uncertainty at a 95% confidence interval as provided in table 1.2 and 1.4 of Chapter 1 of Vol. 2 (Energy) of the 2006 IPCC Guidelines on National GHG Inventories is considered as ex-ante values for the current crediting period for NCV and CO₂ emission factor of diesel.

4.3 Leakage Emissions

As per applied approved methodology no leakage emission relevant to project activity. $LE_y = 0$

4.4 Estimated GHG Emission Reductions and Carbon Dioxide Removals

$$ER_y = BE_y - PE_y - LE_y$$

Where,

ER_y = Emission reductions in year y (t CO₂)

BE_y = Baseline Emissions in year y (t CO₂)

PE_y = Project emissions in year y (t CO₂)

LE_y = Leakage emissions in year y (t CO₂)

Baseline emission is calculated as

$$BE_y = EG_{PJ,y} \times EF_{CM,grid,y}$$

Where,

Estimated Gross electricity generation= 28,840 MWh

Estimated loss and auxiliary consumption= 1%

$EG_{PJ,y} = EG_{PJ, facility,y} = 28,840 * (1-1\%) \text{ MWh (as per FSR)}$

$EG_{PJ, facility,y} = 28,552 \text{ MWh/year}$

$EF_{CM, grid,y} = 0.72205 \text{ tCO}_2\text{e/MWh}$

Hence

$BE_y = 28,552 \times 0.72205 = 20,615 \text{ tCO}_2\text{e/year}$

$BE_y = 20,615 \text{ tCO}_2\text{e/year}$

As crediting period starting from 01st October 2025, the baseline emission for year 2025 (92 days) is calculated as

$BE_{y,2025} = 20,615 \times 92/365 = 5,196 \text{ tCO}_2\text{e}$

Similarly for year 2032

$BE_{y,2032} = 20,615 \times 273/365 = 15,419 \text{ tCO}_2\text{e}$

Project emission:

Ex-ante project emission considered as zero, $PE_y=0$.

Leakage emissions:

No leakage emission as per applied approved methodology, $LE_y=0$.

Vintage period	Estimated baseline emissions (tCO ₂ e)	Estimated project emissions (tCO ₂ e)	Estimated leakage emissions (tCO ₂ e)	Estimated reduction VCUs (tCO ₂ e)	Estimated removal VCUs (tCO ₂ e)	Estimated total VCUs (tCO ₂ e)
01-Oct-2025 to 31-Dec-2025	5,196	0	0	5,196	0	5,196
01-Jan-2026 to 31-Dec-2026	20,615	0	0	20,615	0	20,615
01-Jan-2027 to 31-Dec-2027	20,615	0	0	20,615	0	20,615
01-Jan-2028 to 31-Dec-2028	20,615	0	0	20,615	0	20,615
01-Jan-2029 to 31-Dec-2029	20,615	0	0	20,615	0	20,615
01-Jan-2030 to 31-Dec-2030	20,615	0	0	20,615	0	20,615

01-Jan-2031 to 31-Dec-2031	20,615	0	0	20,615	0	20,615
01-Jan-2032 to 30-Sep-2032	15,419	0	0	15,419	0	15,419
Total	144,305	0	0	144,305	0	144,305

5 MONITORING

5.1 Data and Parameters Available at Validation

Data / Parameter	EF_{diesel}
Data unit	tCO ₂ /TJ
Description	Emission factor of diesel
Source of data	IPCC, 2006 guidelines Table 1.4 of Chapter 1, Volume 2 Reference : IPCC Link
Value applied	74.8
Justification of choice of data or description of measurement methods and procedures applied	The value has been taken from IPCC guidelines for default values, a reliable and conservative source.
Purpose of data	Calculation of project emissions
Comments	The value is fixed ex-ante

Data / Parameter	D_{diesel}
Data unit	Kg/liter
Description	Density of diesel
Source of data	IPCC, 2006 guidelines Table 1.4 of Chapter 1, Volume 2 Reference : IPCC Link
Value applied	0.83

Justification of choice of data or description of measurement methods and procedures applied	The value has been taken from IPCC guidelines for default values, a reliable and conservative source.
Purpose of data	Calculation of project emissions
Comments	The value is fixed ex-ante

Data / Parameter	$EF_{grid,OM,y}$
Data unit	tCO ₂ /MWh
Description	Operating margin CO ₂ emission factor for the project electricity system in year y
Source of data	Calculated
Value applied	0.9239
Justification of choice of data or description of measurement methods and procedures applied	Calculated in line with “<i>Tool to calculate the emission factor for an electricity system (v07.0)</i>” using data from “Research and develop emission factor (EF) of Vietnam's electricity grid in 2021 (attached with CV 1278/BDKH-TTBVTOD)” published dated 21/12/2022 The value used is calculated ex-ante as generation based weighted average of last three years of the operating margin provided in the “Research and develop emission factor (EF) of Vietnam's electricity grid in 2021 (attached with CV 1278/BDKH-TTBVTOD)”.
Purpose of data	For calculation of baseline emission
Comments	This value is fixed ex-ante

Data / Parameter	$EF_{grid,BM,y}$
Data unit	tCO ₂ /MWh
Description	Build margin CO ₂ emission factor for the project electricity system in year y
Source of data	Calculated
Value applied	0.5202
Justification of choice of data or description of measurement methods and procedures applied	Calculated in line with “ <i>Tool to calculate the emission factor for an electricity system (v07.0)</i> ” using data from “Research and develop

	emission factor (EF) of Vietnam's electricity grid in 2021 (attached with CV 1278/BDKH-TTBVTOD)" published dated 21/12/2022
	The value is calculated ex-ante as most recent build margin provided by the VEN.
Purpose of data	For calculation of baseline emission
Comments	This value is fixed ex-ante

Data / Parameter	$EF_{grid,CM,y} / EF_{grid,y}$
Data unit	tCO ₂ /MWh
Description	Combined margin CO ₂ emission factor for the project electricity system in year y
Source of data	Calculated
Value applied	0.72205
Justification of choice of data or description of measurement methods and procedures applied	This has been calculated based on Operating Margin (OM) and Build Margin (BM) using data from "Research and develop emission factor (EF) of Vietnam's electricity grid in 2021 (attached with CV 1278/BDKH-TTBVTOD)" published dated 21/12/2022. Please refer section B.6.1 for details
Purpose of data	For calculation of baseline emission
Comments	This value is fixed ex-ante

5.2 Data and Parameters Monitored

Data / Parameter	$EG_{PJ,facility,y}$
Data unit	MWh
Description	Quantity of net electricity generation supplied by the project plant/unit to the grid in year y
Source of data	Monthly readings report
Description of measurement methods and procedures to be applied	Calculated based on energy export and import recorded in monthly meter readings from electricity meters.

Frequency of monitoring/recording	Continuous monitoring, monthly recording
Value applied	28,552
Monitoring equipment	Not applicable
QA/QC procedures to be applied	NA
Purpose of data	Calculation of baseline emissions
Calculation method	This parameter is calculated based on monitored values from bi-directional energy meters. The following method to calculate the net electricity supplied to grid (a) The quantity of electricity supplied by the project plant/unit to the grid; and (b) The quantity of electricity delivered to the project plant/unit from the grid $EG_{PJ, facility, y} = EG_{PJ, export, y} - EG_{PJ, import, y}$
Comments	NA

Data / Parameter	EG_{Export,y}
Data unit	MWh
Description	Quantity of total electricity generation supplied by the project plant/unit to the grid in year y
Source of data	Monthly readings report
Description of measurement methods and procedures to be applied	The electricity exported to grid is monitored using energy meters installed at project site and at substation.
Frequency of monitoring/recording	Continuous monitoring, monthly recording
Value applied	36,530
Monitoring equipment	Tri-vector energy meter Accuracy class: 0.2s
QA/QC procedures to be applied	According to the recommendation by the manufacturer or the regulations of the grid company, meters will be calibrated periodically at least once in 3 years. Data measured by meters will be cross-checked with the sales record.

Purpose of data	Calculation of baseline emissions
Calculation method	This parameter will be monitored using bi-directional energy meter.
Comments	--

Data / Parameter	$EG_{PJ,Import,y}$
Data unit	MWh
Description	Quantity of electricity supplied to the project plant/unit by the grid in year y
Source of data	Monthly reading report
Description of measurement methods and procedures to be applied	The electricity imported to grid is monitored using energy meters installed at project site and at substation.
Frequency of monitoring/recording	Continuous monitoring, monthly recording
Value applied	0 for ex-ante calculations
Monitoring equipment	Tri-vector electricity meter Accuracy class: 0.2s
QA/QC procedures to be applied	According to the recommendation by the manufacturer or the regulations of the grid company, meters will be calibrated periodically at least once in 3 years. Data measured by meters will be cross-checked with the sales record.
Purpose of data	Calculation of baseline emissions
Calculation method	This parameter will be monitored using bi-directional energy meter.
Comments	--

Data / Parameter	$FC_{I,y}$
Data unit	Liter/year
Description	Quantity of diesel used in year y
Source of data	Plant log books

Description of measurement methods and procedures to be applied	The value will be recorded based on total purchase and remaining stock.
Frequency of monitoring/recording	Each filling, Monthly recording
Value applied	0 for ex-ante
Monitoring equipment	Volume measurement
QA/QC procedures to be applied	The purchase and stock will be matched to check total usage
Purpose of data	Calculation of project emissions
Calculation method	Monitored
Comments	Data will be archived for crediting period +2 years

5.3 Monitoring Plan

This monitoring plan is developed in accordance with the modalities and procedures for VCS project activity and in line with the guidance provided in the applied methodology. The monitoring plan, which will be implemented by the project proponent, describes about overall monitoring organization, Internal reporting procedure for data collection and archiving, Calibration of measurement equipment's, Training procedures and internal audit procedures.

Data and parameters, which will be monitored under this monitoring plan as sited in monitoring plan will be measured and strictly monitored at the project site by means of accurately calibrated instruments.

The monitoring plan structure and the roles of the different member in the monitoring plan are explained below.

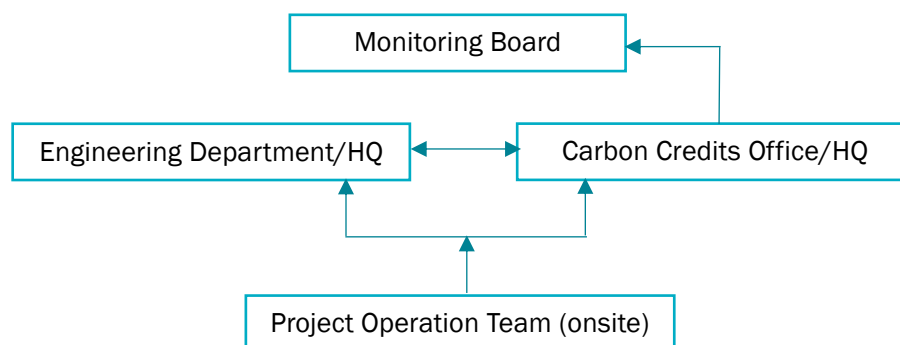
Monitoring Organization:

Technicians and operators are responsible for the recording of the relevant data in plant log books. Shift in-charges are also responsible of QA/QC of data recorded and maintenance of log-books and other records.

These records in plant log books will consolidate in the form of monthly GHG report by plant supervisor and will be submitted to plant manager for review. The Management, who represents the overall responsibility on the project activity, will be able, with this operational structure, to monitor the development of the project activity and make the relevant amendments to the process.

The VCS monitoring team will compose the following staff:

<u>Position</u>	<u>Report to:</u>
Operators	Site Engineer
Site Engineer	Site In-charge
Site In-charge	Engineering Department/Carbon Credit office
Carbon Credit office	Monitoring Board



Monitoring Provisions:

The power generated by the three 3.4MW generators will be supplied to transformer substation through a 115 kV single-circuit transmission line to EVN Grid. According to the Power Purchase Agreement, two meters will be installed to monitor the input/output power. There will be two meters i.e. main meter and back up meter, installed at the grid access points, to monitoring the input/output electricity at the grid side. Both the main and backup meters will be installed in parallel combination. When there is anything wrong with the main meter, the backup meter will be adopted. The accuracy of all meters will be not less than 0.5s.

Data collection:

The specific steps for data collection and reporting are listed below:

- During the crediting period, both the grid company and the proposed project owner will record the values displayed by the main meter.
- Simultaneously to step a), the proposed VCS project owner will both record the values displayed by the backup meters.
- The meters will be calibrated according to the relevant regulation and request of grid.
- The main meter's readings will be cross-checked with record document confirmed by grid.
- The proposed VCS project owner and the grid company will record both output and input

power readings from the main meter. These data will be used to calculate the amount of net electricity delivered to the grid.

f) The proposed VCS project owner will be responsible of providing copies of record document confirmed by PP to the VVB for verification.

If the reading of the main meter in a certain month is inaccurate and beyond the allowable error or the meter doesn't work normally, the grid-connected power generation shall be determined by following measures:

a) Read the data of the backup meters.

b) If the backup meter's data is not so accurate as to be accepted, or the practice is not standardized, the proposed VCS project owner and the grid corporation should jointly make a reasonable and conservative estimation method which can be supported by sufficient evidence and proved to be reasonable and conservative when verified by Verifier.

c) If the proposed VCS project owner and the grid corporation don't agree on an estimated method, arbitration will be conducted according the procedures set by the agreement to work out an estimation method.

Calibration:

Calibration of Meters & Metering should be implemented according to relevant standards and rules accepted by the grid company. After the examination, the meters should be sealed. The meters shall be calibrated at least once in 3 years by grid. The lift of the seals requires the presence of both the proposed VCS project owner and the grid company. One party must not lift the seals or fiddle with the meters without the presence of the other party.

All the meters installed shall be tested and calibrated by a qualified metering verification institution commissioned jointly by the proposed VCS project owner and the grid company after:

- 1) Detection of a difference larger than the allowable error in the readings of both meters;
- 2) The repair of all or part of meter caused by the failure of one or more parts to operate in accordance with the specifications.

Training and Maintenance:

All persons that form the part of the VCS Project Management Team shall receive appropriate training which shall provide an overview of the VCS and cover all elements of monitoring plan in detail. Training for operating and maintaining the plant shall also be provided.

Emergency Preparedness:

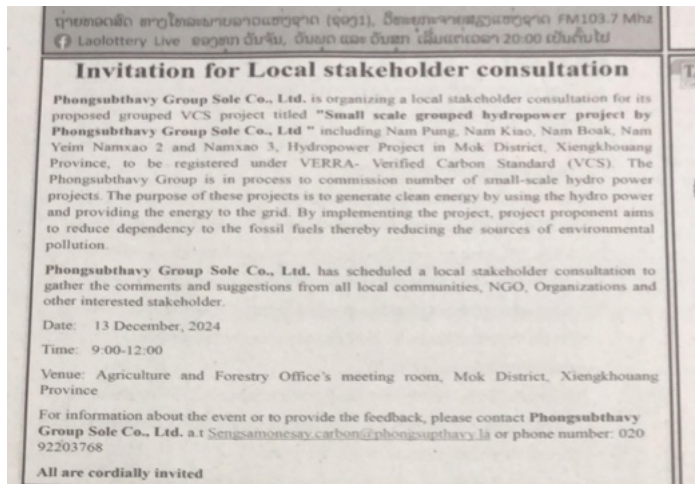
Being a small hydro plant intensive emergency preparation is not needed, however, the following may be highlighted:

- There are no critical loads that may require emergency power supply in case of total outage.
- A fire extinguisher has been installed in case of small fires
- Information regarding any safety earthing systems in place.

APPENDIX 1: COMMERCIALY SENSITIVE INFORMATION

Section	Information	Justification
<u>NA</u>	NA	NA

APPENDIX 2: LOCAL STAKEHOLDER CONSULTATION



[illegible]

ລ/ດ No	ຊື່ ແລະ ນາມສະກຸນ Name and Surname	ຕຳແໜ່ງ Position	ຜກສ່ວນ Organization	ຂໍ້ມູນຕິດຕໍ່ Contact Information	ລາຍຊື່ Signature
1	ທ ວິໄລພອນ ພະມິວົງ	ເຈົ້າໜີ້ສັງຄົມ ພະມິວົງ	ບ/ບ ກຸ່ມປະຊາກອນ ພະມິວົງ	29998898	
2	ນ. ພອນ ພະມິວົງ	ອົງການ	ກຸ່ມປະຊາກອນ ພະມິວົງ	22219986	
3	ທ. ພິມວົງ ພະມິວົງ	ອົງການ	ກຸ່ມປະຊາກອນ ພະມິວົງ	22081508	
4	ທ ພິມວົງ ພະມິວົງ	ອົງການ	ກຸ່ມປະຊາກອນ ພະມິວົງ	95457406	
5	ທ. ພິມວົງ ພະມິວົງ	ອົງການ	ກຸ່ມປະຊາກອນ ພະມິວົງ	95663852	
6	ທ ພິມວົງ ພະມິວົງ	ອົງການ	ກຸ່ມປະຊາກອນ ພະມິວົງ	99777288	
7	ທ ພິມວົງ ພະມິວົງ	ອົງການ	ກຸ່ມປະຊາກອນ ພະມິວົງ	22344732	
8	ທ ພິມວົງ ພະມິວົງ	ອົງການ	ກຸ່ມປະຊາກອນ ພະມິວົງ	5599599	

ລ/ດ No	ຊື່ ແລະ ນາມສະກຸນ Name and Surname	ຕຳແໜ່ງ Position	ພາກສ່ວນ Organization	ຂໍ້ມູນຕິດຕໍ່ Contact Information	ລາຍເຊັນ Signature
9	ທ. ສິນທິ ສິນທິ	ວິຊາການ	ໜີ້ສິດ. ຂ.	໒໒741100	
10	ທ. ສາວສິນທິ	ບຸກຄົນ	ໜີ້ສິດ	໑໑873020	
11	ທ. ສິນທິ ສິນທິ	ໂຄສະນາ	ໜີ້ສິດ	໒3831400	
12	ທ. ສິນທິ ສິນທິ	ບຸກຄົນ	ໜີ້ສິດ	໒354 ໑738	
13	ທ. ສິນທິ	ບຸກຄົນ	ໜີ້ສິດ	໑໑386922	
14	ທ. ສິນທິ ສິນທິ	ໜີ້ສິດ	ໜີ້ສິດ	໒໒໑໑໑738	
15	ທ. ສິນທິ ສິນທິ	ບຸກຄົນ	ໜີ້ສິດ	໑໑7໒79໑໑	
16	ທ. ສິນທິ	ໜີ້ສິດ	ໜີ້ສິດ	໑໑໑໑໑໑໑໑	
17	ທ. ສິນທິ	ໜີ້ສິດ	ໜີ້ສິດ	໒໒໒໒໒໒໒໒	
18	ທ. ສິນທິ ສິນທິ	ບຸກຄົນ	ໜີ້ສິດ	໒໒໒໒໒໒໒໒	
19	ທ. ສິນທິ ສິນທິ	ໜີ້ສິດ	ໜີ້ສິດ	໒໒໒໒໒໒໒໒	
20	ທ. ສິນທິ ສິນທິ	ໜີ້ສິດ	ໜີ້ສິດ	໒໒໒໒໒໒໒໒	
21	ທ. ສິນທິ ສິນທິ	ໜີ້ສິດ	ໜີ້ສິດ	໒໒໒໒໒໒໒໒	

ឈ្មោះ No	ឈ្មោះ ឈ្មោះ Name and Surname	តំណែង Position	ស្ថាប័ន Organization	ព័ត៌មាន Contact Information	ហត្ថលេខា Signature
22	ឈ. ឈុន ឈុន	ឧបនាយក	ឧត្តម - ឈុន	22340579	
23	ឈ. ឈុន ឈុន	ឧបនាយក	ឧត្តម - ឈុន	29992453	
24	ឈ. ឈុន ឈុន	ឧបនាយក	ឧត្តម - ឈុន	9507500	
25	ឈ. ឈុន ឈុន	ឧបនាយក	ឧត្តម - ឈុន	0022426800	
26	ឈ. ឈុន ឈុន	ឧបនាយក	ឧត្តម - ឈុន	02024546505	
27	ឈ. ឈុន ឈុន	ឧបនាយក	ឧត្តម - ឈុន	23529559	
28	ឈ. ឈុន ឈុន	ឧបនាយក	ឧត្តម - ឈុន	56427545	
29	ឈ. ឈុន ឈុន	ឧបនាយក	ឧត្តម - ឈុន	54091891	
30	ឈ. ឈុន ឈុន	ឧបនាយក	ឧត្តម - ឈុន	92111616	
31	ឈ. ឈុន ឈុន	ឧបនាយក	ឧត្តម - ឈុន	55301935	
32	ឈ. ឈុន ឈុន	ឧបនាយក	ឧត្តម - ឈុន	22229181	
33	ឈ. ឈុន ឈុន	ឧបនាយក	ឧត្តម - ឈុន	52080005	
34	ឈ. ឈុន ឈុន	ឧបនាយក	ឧត្តម - ឈុន	28211112	

ឈ្មោះ No	ឈ្មោះ ឈ្មោះ Name and Surname	តំណែង Position	ស្ថាប័ន Organization	ព័ត៌មាន Contact Information	ហត្ថលេខា Signature
35	ឈ. ឈុន ឈុន	ឧបនាយក	ឧត្តម - ឈុន	0309219380	
36	ឈ. ឈុន ឈុន	ឧបនាយក	ឧត្តម - ឈុន	59599165	
37	ឈ. ឈុន ឈុន	ឧបនាយក	ឧត្តម - ឈុន	56982599	
38	ឈ. ឈុន ឈុន	ឧបនាយក	ឧត្តម - ឈុន	29155559	
39	ឈ. ឈុន ឈុន	ឧបនាយក	ឧត្តម - ឈុន	03044970354	
40	ឈ. ឈុន ឈុន	ឧបនាយក	ឧត្តម - ឈុន	29551132	
41	ឈ. ឈុន ឈុន	ឧបនាយក	ឧត្តម - ឈុន	28561990	
42	ឈ. ឈុន ឈុន	ឧបនាយក	ឧត្តម - ឈុន	225333221	
43	ឈ. ឈុន ឈុន	ឧបនាយក	ឧត្តម - ឈុន	0309481233	
44	ឈ. ឈុន ឈុន	ឧបនាយក	ឧត្តម - ឈុន	95540902	
45	ឈ. ឈុន ឈុន	ឧបនាយក	ឧត្តម - ឈុន	22533030	
46	ឈ. ឈុន ឈុន	ឧបនាយក	ឧត្តម - ឈុន	54664636	
47	ឈ. ឈុន ឈុន	ឧបនាយក	ឧត្តម - ឈុន	08491829	

ឈ្មោះ No	ឈ្មោះ ឈ្មោះ Name and Surname	តំណែង Position	ស្ថាប័ន Organization	ព័ត៌មាន Contact Information	ហត្ថលេខា Signature
48	ឈ. ឈុន ឈុន	ឧបនាយក	ឧត្តម - ឈុន	55374006	
49	ឈ. ឈុន ឈុន	ឧបនាយក	ឧត្តម - ឈុន	02092203768	
50	ឈ. ឈុន ឈុន	ឧបនាយក	ឧត្តម - ឈុន	02053523844	
51	ឈ. ឈុន ឈុន	ឧបនាយក	ឧត្តម - ឈុន	56202992	
52					